



Database Systems Final Project

Department: BS-SE (Section B)

Submission Date: 4th May, 2026

Submitted To: Prof Zoya Sumbul

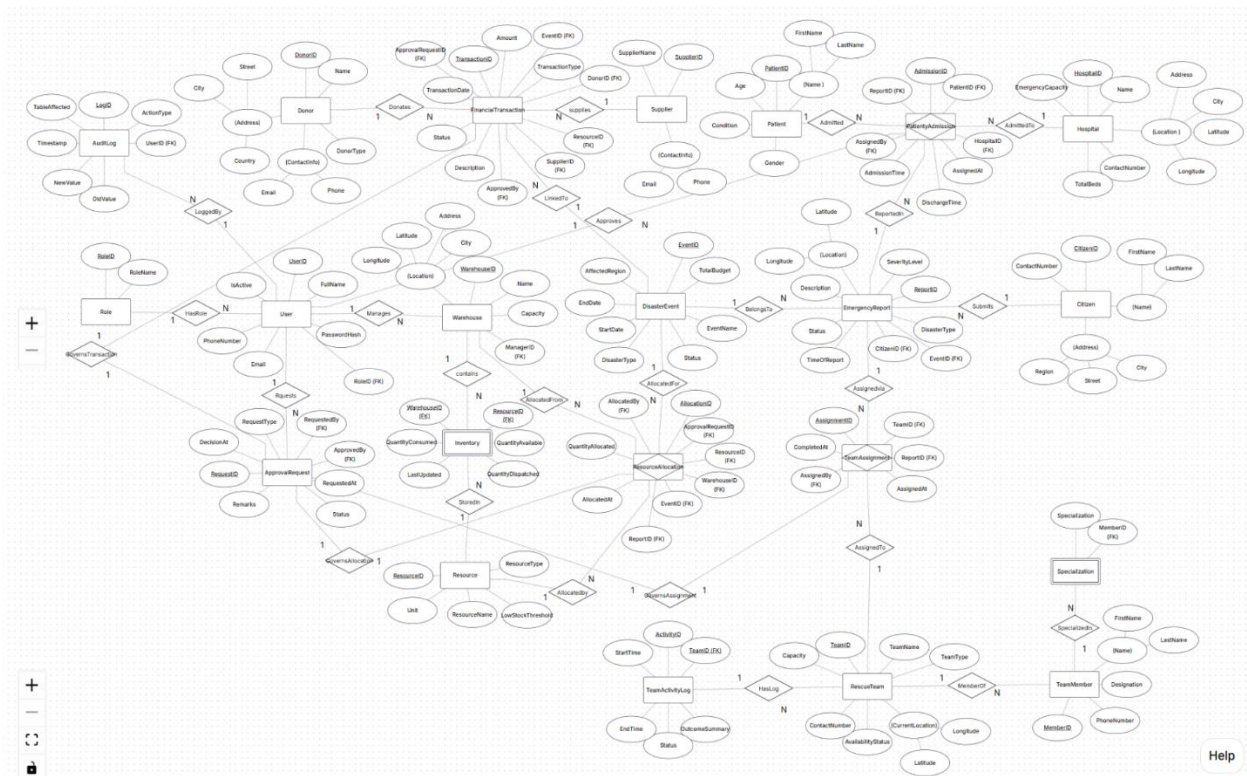
Submitted By

Student Name	Student ID
Saliha Noor	24I-3066
Sajal Ishtiaq	24I-3041

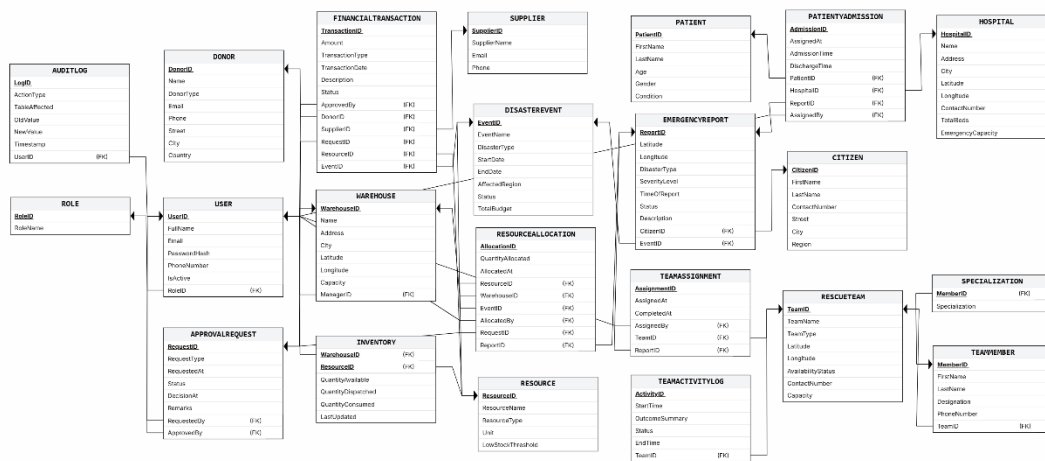
Contents

1. Entity Relation Diagram	3
2. Relational Schema	3
3. Normalization Steps.....	4
4. SQL Implementation (DDL, DML, Queries):	9
5. Transaction Handling Demonstration	17
6. Trigger Implementation & Use Cases.....	19
7. View Definitions & Performance Comparison (Latency Analysis).....	20
8. Indexing Strategy & Query Performance Report.....	25
9. Frontend Interface.....	36
10. Design Rationale	44
11. MIS Reports and Dashboards	46
12. System Demonstration	48

1. Entity Relation Diagram



2. Relational Schema



3. Normalization Steps

Step 1: Unnormalized Form (UNF)

Initially, all system data was stored in a single flat relation containing mixed and unstructured attributes.

Included Issues:

- Repeating groups (multiple resources, reports, team members)
- Multi-valued attributes (e.g., specializations, contact info)
- Nested/composite data (patient, inventory details)
- Derived attributes (e.g., available beds)

Example:

```
DISASTER_SYSTEM(  
  UserID, FullName, Email, RoleName, Permissions,  
  DonorID, DonorName, SupplierID,  
  TransactionID, Amount, TransactionType,  
  WarehouseID, ResourceID, QuantityAvailable,  
  EventID, DisasterType,  
  ReportID, SeverityLevel,  
  HospitalID, TotalBeds,  
  PatientID, PatientDetails,  
  TeamID, TeamType,  
  MemberID, Specialization,  
  LogID, ActionType  
)
```

Problems:

- Data redundancy
- Update, insertion, deletion anomalies
- Non-atomic attributes

Step 2: First Normal Form (1NF)

To achieve 1NF, the large flat relation was decomposed into separate tables. All multi-valued attributes were eliminated, each attribute was made atomic, and a primary key was assigned to every table.

Key Changes:

- Split name and address fields
- Created **SPECIALIZATION** table for multi-valued attributes

Steps performed to achieve 1NF:

- Eliminated all repeating groups by creating separate tables
- Ensured every attribute is atomic, no multi-valued or composite fields
- Separated nested structures (e.g., patient details out of disaster records)
- Assigned a primary key to each new table
- Separated name fields: FirstName and LastName stored separately (not as one 'Name' field)
- Separated address fields: Street, City, Region stored separately (not as one 'Address' field)
- Created a separate SPECIALIZATION table to handle the multi-valued specialization attribute of team members

Example Decomposition:

I. ROLE(RoleID, RoleName)

II. USER(UserID, FullName, Email, PasswordHash, RoleID)

III. DONOR(DonorID, Name, Email, Phone, Address)

IV. SUPPLIER(SupplierID, SupplierName, Email, Phone)

V. RESOURCE(ResourceID, ResourceName, ResourceType, Unit)

VI. WAREHOUSE(WarehouseID, Name, Location, Capacity)

VII. INVENTORY(WarehouseID, ResourceID, QuantityAvailable)

VIII. EMERGENCYREPORT(ReportID, Location, Severity, CitizenID)

IX. TEAMMEMBER(MemberID, FirstName, LastName, TeamID)

X. SPECIALIZATION(MemberID, Specialization)

Step 3: Second Normal Form (2NF)

To achieve 2NF, all partial dependencies were eliminated. A partial dependency occurs when a non-key attribute depends on only **part** of a composite primary key rather than the entire key. This is only relevant for tables with composite primary keys.

Issues Found:

- Composite keys in **INVENTORY** and **SPECIALIZATION**

Steps performed to achieve 2NF:

- Confirmed all tables are already in 1NF
- Identified all tables with composite primary keys
- Checked every non-key attribute for partial dependency on any subset of the composite key
- Moved partially dependent attributes into their own separate table

Examples:

Before	After	Problem
INVENTORY(WarehouseID, ResourceID, ResourceName, Quantity)	RESOURCE(ResourceID, ResourceName, Type, Unit) + INVENTORY(WarehouseID, ResourceID, Quantity)	ResourceName depends only on ResourceID
SPECIALIZATION(MemberID, Specialization, MemberName)	TEAMMEMBER(MemberID, Name) + SPECIALIZATION(MemberID, Specialization)	MemberName depends only on MemberID

Step 4: Third Normal Form (3NF)

To achieve 3NF, all transitive dependencies were eliminated. A transitive dependency occurs when a non-key attribute depends on another non-key attribute rather than directly on the primary key.

Steps performed to achieve 3NF:

- Confirmed all tables are already in 2NF
- Identified non-key attributes that could potentially determine other non-key attributes
- Extracted transitively dependent attributes into their own tables

Replaced repeated data with foreign key references

Issues Found:

Before	After	Problem
--------	-------	---------

USER(UserID, RoleName, Permissions)	USER(UserID, RoleID) + ROLE(RoleID, RoleName)	Permissions depend on Role
FINANCIALTRANSACTION(TransactionID, DonorName)	DONOR(DonorID, Name) + FINANCIALTRANSACTION(TransactionID, DonorID)	DonorName depends on DonorID
WAREHOUSE(WarehouseID, ManagerName)	USER(UserID, Name) + WAREHOUSE(WarehouseID, ManagerID)	Manager depends on User
TEAMASSIGNMENT(AssignmentID, TeamName)	RESCUETEAM(TeamID) + TEAMASSIGNMENT(AssignmentID, TeamID)	TeamName depends on TeamID

Step 5: Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form (BCNF) is a stricter version of Third Normal Form (3NF). A relation is in BCNF if:

- For every non-trivial functional dependency ($X \rightarrow Y$), X must be a **superkey**
- A superkey is an attribute or set of attributes that can uniquely identify every row

Even after achieving 3NF, anomalies may still exist when there are overlapping candidate keys or when a non-key attribute determines part of a composite key. BCNF addresses these remaining cases.

Before (Violates BCNF)		After (BCNF Compliant)
Structure	USER(UserID [PK], FullName, RoleName, Permission)	ROLE(RoleID [PK], RoleName) USER(UserID [PK], FullName, Email, PasswordHash, IsActive, RoleID [FK->ROLE])
FDs	UserID $\rightarrow$ RoleName UserID $\rightarrow$ Permission RoleName $\rightarrow$ Permission (RoleName is not a superkey — BCNF violation)	UserID $\rightarrow$ RoleID (UserID is superkey — OK) RoleID $\rightarrow$ RoleName (RoleID is superkey of ROLE — OK) All FDs have superkeys on the left side

Why fixed	RoleName determining Permission is a BCNF violation because RoleName alone cannot uniquely identify a user	<i>ROLE is its own table with RoleID as PK. Every FD now has a superkey on the left. BCNF satisfied.</i>
------------------	--	--

Final Result:

- All tables satisfy BCNF
- Composite key tables (INVENTORY, SPECIALIZATION) verified
- No remaining anomalies

#	Table	Final BCNF Schema
1	ROLE	<i>ROLE(RoleID [PK], RoleName)</i>
2	USER	<i>USER(UserID [PK], FullName, Email, PasswordHash, PhoneNumber, IsActive, RoleID [FK])</i>
3	AUDITLOG	<i>AUDITLOG(LogID [PK], ActionType, TableAffected, OldValue, NewValue, Timestamp, UserID [FK])</i>
4	DONOR	<i>DONOR(DonorID [PK], Name, DonorType, Email, Phone, Street, City, Country)</i>
5	SUPPLIER	<i>SUPPLIER(SupplierID [PK], SupplierName, Email, Phone)</i>
6	FINANCIALTRANSACTION	<i>FINANCIALTRANSACTION(TransactionID [PK], Amount, TransactionType, TransactionDate, Description, Status, ApprovedBy [FK], DonorID [FK], SupplierID [FK], RequestID [FK], ResourceID [FK], EventID [FK])</i>
7	APPROVALREQUEST	<i>APPROVALREQUEST(RequestID [PK], RequestType, RequestedAt, Status, DecisionAt, Remarks, RequestedBy [FK], ApprovedBy [FK])</i>
8	WAREHOUSE	<i>WAREHOUSE(WarehouseID [PK], Name, Address, City, Latitude, Longitude, Capacity, ManagerID [FK])</i>
9	RESOURCE	<i>RESOURCE(ResourceID [PK], ResourceName, ResourceType, Unit, LowStockThreshold)</i>
10	INVENTORY	<i>INVENTORY({WarehouseID, ResourceID} [Composite PK], QuantityAvailable, QuantityDispatched, QuantityConsumed, LastUpdated)</i>
11	RESOURCEALLOCATION	<i>RESOURCEALLOCATION(AllocationID [PK], QuantityAllocated, AllocatedAt, ResourceID [FK], WarehouseID [FK], EventID [FK], AllocatedBy [FK], RequestID [FK], ReportID [FK])</i>
12	DISASTEREVENT	<i>DISASTEREVENT(EventID [PK], EventName, DisasterType, StartDate, EndDate, AffectedRegion, Status, TotalBudget)</i>
13	CITIZEN	<i>CITIZEN(CitizenID [PK], FirstName, LastName, ContactNumber, Street, City, Region)</i>

14	EMERGENCYREPORT	<i>EMERGENCYREPORT</i> (ReportID [PK], Latitude, Longitude, DisasterType, SeverityLevel, TimeOfReport, Status, Description, CitizenID [FK], EventID [FK])
15	HOSPITAL	<i>HOSPITAL</i> (HospitalID [PK], Name, Address, City, Latitude, Longitude, ContactNumber, TotalBeds, EmergencyCapacity)
16	PATIENT	<i>PATIENT</i> (PatientID [PK], FirstName, LastName, Age, Gender, Condition)
17	PATIENTADMISSION	<i>PATIENTADMISSION</i> (AdmissionID [PK], AdmissionTime, DischargeTime, AssignedAt, PatientID [FK], HospitalID [FK], ReportID [FK], AssignedBy [FK])
18	RESCUETEAM	<i>RESCUETEAM</i> (TeamID [PK], TeamName, TeamType, Latitude, Longitude, AvailabilityStatus, ContactNumber, Capacity)
19	TEAMMEMBER	<i>TEAMMEMBER</i> (MemberID [PK], FirstName, LastName, Designation, PhoneNumber, TeamID [FK])
20	SPECIALIZATION	<i>SPECIALIZATION</i> (MemberID, Specialization) [Composite PK], MemberID [FK])
21	TEAMASSIGNMENT	<i>TEAMASSIGNMENT</i> (AssignmentID [PK], AssignedAt, CompletedAt, AssignedBy [FK], TeamID [FK], ReportID [FK])
22	TEAMACTIVITYLOG	<i>TEAMACTIVITYLOG</i> (ActivityID [PK], StartTime, OutcomeSummary, Status, EndTime, TeamID [FK])

4. SQL Implementation (DDL, DML, Queries)

DDL:

```

1  CREATE DATABASE DisasterMIS;
2
3  USE DisasterMIS;
4
5  -- DDL - TABLE CREATION
6
7  -- 1.1 ROLE
8  CREATE TABLE ROLE (
9      RoleID INT NOT NULL IDENTITY(1,1),
10     RoleName VARCHAR(50) NOT NULL,
11     CONSTRAINT PK_ROLE PRIMARY KEY (RoleID),
12     CONSTRAINT UQ_ROLE_Name UNIQUE (RoleName)
13 );
14

```

81 % 1 0 Ln: 1, Ch: 29 (28 chars, 1 lines) SPC CRLF UTF-8

Messages

Commands completed successfully.

Completion time: 2026-05-04T00:14:40.8061098+05:00

```
Deliverable4...(\Sajal (54))*
270 -- 1.21 SUPPLIER
271 CREATE TABLE SUPPLIER (
272     SupplierID INT NOT NULL IDENTITY(1,1),
273     SupplierName VARCHAR(150) NOT NULL,
274     Email VARCHAR(150) NULL,
275     Phone VARCHAR(20) NULL,
276     CONSTRAINT PK_SUPPLIER PRIMARY KEY (SupplierID)
277 );
278
279 -- 1.22 FINANCIALTRANSACTION
280 CREATE TABLE FINANCIALTRANSACTION (
281     TransactionID INT NOT NULL IDENTITY(1,1),
282     Amount DECIMAL(15,2) NOT NULL,
283     TransactionType VARCHAR(50) NOT NULL,
284 );
```

81 % No issues found Ln: 300, Ch: 3 (12455 chars, 295 lines) SPC CRLF UTF-8

Messages

Commands completed successfully.

Completion time: 2026-05-04T00:15:11.0004875+05:00

DML:

```
Deliverable4...(\Sajal (54))*
485 ('Herbion Pharma', 'orders@herbion.com', '042-35976000'),
486 ('Metro Cash and Carry', 'corporate@metro.com.pk', '021-32456789'),
487 ('PSO Fuel Services', 'fuel@psso.com.pk', '021-32068000');
488
489 -- 2.21 FINANCIAL TRANSACTIONS
490 INSERT INTO FINANCIALTRANSACTION
491 (Amount, TransactionType, TransactionDate, Description, Status, ApprovedBy, DonorID, SupplierID, RequestID, ResourceID)
492 VALUES
493 (500000.00, 'Donation', '2025-07-05 10:00:00', 'Flood relief donation from Hashoo', 'Approved', 1, 1, NULL, NULL, NULL),
494 (250000.00, 'Donation', '2025-07-06 11:00:00', 'Donation from USAID for Karachi Flood', 'Approved', 1, 4, NULL, NULL, NULL),
495 (180000.00, 'Expense', '2025-07-07 09:00:00', 'Procurement of food supplies', 'Approved', 5, NULL, NULL, NULL, NULL),
496 (95000.00, 'Expense', '2025-03-16 08:00:00', 'Medical supplies for earthquake', 'Approved', 5, NULL, NULL, NULL, NULL),
497 (320000.00, 'Donation', '2025-03-17 12:00:00', 'Alkhidmat donation for earthquake relief', 'Approved', 1, 2, NULL, NULL, NULL),
498 (75000.00, 'Procurement', '2025-09-12 10:00:00', 'Fire-fighting equipment purchase', 'Pending', NULL, NULL, NULL, NULL, NULL);
```

81 % No issues found Ln: 501, Ch: 5 (13049 chars, 198 lines) SPC CRLF UTF-8

Messages

(5 rows affected)

(7 rows affected)

(7 rows affected)

(5 rows affected)

(7 rows affected)

(5 rows affected)

81 % No issues found Ln: 45, Ch: 1 TABS MIXED UTF-8 with BOM

Queries:

```
1079
1080 -- SECTION 7: MIS REPORTING QUERIES
1081
1082 -- Q1: Incident count by DisasterType and SeverityLevel
1083 SELECT
1084     DisasterType,
1085     SeverityLevel,
1086     COUNT(*) AS TotalReports
1087 FROM EMERGENCYREPORT
1088 GROUP BY DisasterType, SeverityLevel
1089 ORDER BY TotalReports DESC;
1090
1091 -- Q2: Resource utilisation per warehouse (via view)
1092 SELECT *
```

89 % No issues found

Results Messages

	DisasterType	SeverityLevel	TotalReports
1	Flood	Critical	2
2	Heatwave	Low	1
3	Landslide	High	1
4	Earthquake	High	1
5	Fire	High	1
6	Fire	Medium	1

1090

1091

1092

1093

1094

1095

```
-- Q2: Resource utilisation per warehouse (via view)
SELECT *
FROM vw_inventory_summary
ORDER BY StockStatus DESC, WarehouseName;
```

89 %

No issues found

Ln: 1095, Ch: 1 | (80 chars, 3 lines) | SPC | CRLF | UTF-8

Results

Messages

	WarehouseID	WarehouseName	City	ResourceID	ResourceName	ResourceType	Unit	QuantityAvailable	QuantityDispatched	QuantityCon
1	3	Islamabad Reserve	Islamabad	6	ORS Sachets	Medicine	Box	1000	150	100
2	3	Islamabad Reserve	Islamabad	7	Diesel Generator	Equipment	Liter	2000	200	180
3	1	Karachi Central Store	Karachi	1	Rice Bags	Food	Kg	2000	300	200
4	1	Karachi Central Store	Karachi	2	Bottled Water	Water	Liter	5000	500	400
5	1	Karachi Central Store	Karachi	3	Paracetamol 500mg	Medicine	Box	800	100	80
6	1	Karachi Central Store	Karachi	4	Tent 4-person	Shelter	Unit	120	20	15
7	2	Lahore Hub	Lahore	1	Rice Bags	Food	Kg	1500	200	100
8	2	Lahore Hub	Lahore	2	Bottled Water	Water	Liter	3000	300	200
9	2	Lahore Hub	Lahore	5	Blankets	Shelter	Unit	400	50	30
10	5	Peshawar Store	Peshawar	1	Rice Bags	Food	Kg	800	100	80
11	5	Peshawar Store	Peshawar	3	Paracetamol 500mg	Medicine	Box	400	60	40
12	4	Quetta Depot	Quetta	8	First Aid Kit	Medicine	Kit	300	40	30

Deliverable4...Sajal (54)*

1097

1098

1099

1100

1101

1102

1103

1104

1105

1106

1107

1108

1109

1110

```
SELECT
rt.TeamName,
rt.TeamType,
ta.AssignedAt,
ta.EndTime,
DATEDIFF(HOUR, ta.AssignedAt, ta.EndTime) AS ResponseHours,
ta.Status,
ta.OutcomeSummary
FROM TEAMASSIGNMENT ta
JOIN RESCUETEAM rt ON rt.TeamID = ta.TeamID
JOIN TEAMACTIVITYLOG tal ON tal.TeamID = ta.TeamID
ORDER BY ResponseHours;
```

```
-- Q4: Financial summary by event
```

89 %

No issues found

Ln: 1109, Ch: 1 | (342 chars, 12 lines) | SPC | CRLF

Results

Messages

	TeamName	TeamType	AssignedAt	EndTime	ResponseHours	Status	OutcomeSummary
1	Alpha Rescue	Rescue	2025-07-02 09:00:00.000	NULL	NULL	In Progress	NULL
2	Gamma Fire	Fire	2025-09-11 13:30:00.000	NULL	NULL	In Progress	NULL
3	Delta Search	Rescue	2025-08-21 06:30:00.000	2025-08-21 12:30:00.000	6	Completed	Road cleared after 6 hours
4	Epsilon Med	Medical	2025-06-02 11:30:00.000	2025-06-02 22:00:00.000	11	Completed	80 patients treated for heatstroke
5	Beta Medical	Medical	2025-03-15 05:00:00.000	2025-03-20 18:00:00.000	133	Completed	150 survivors evacuated

```

1109
1110 -- Q4: Financial summary by event
1111 SELECT * FROM vw_event_budget_analysis
1112 ORDER BY TotalExpenses DESC;
1113
1114 -- Q5: Top donors by total approved donations
1115 SELECT
1116     d.Name,
1117     d.DonorType,
1118     SUM(t.Amount) AS TotalDonated
1119 FROM FINANCIALTRANSACTION t
1120 JOIN DONOR d ON d.DonorID = t.DonorID
1121 WHERE t.TransactionType = 'Donation'
1122 AND t.Status = 'Approved'

```

89 %

✓ No issues found

Ln: 1112, Ch: 29 (68 chars, 2 lines)

Results Messages

	EventID	EventName	DisasterType	Status	TotalBudget	TotalDonations	TotalExpenses	RemainingBudget
1	1	Karachi Flood 2025	Flood	Active	5000000.00	750000.00	300000.00	4700000.00
2	2	Balochistan Quake	Earthquake	Resolved	3000000.00	320000.00	95000.00	2905000.00
3	3	Lahore Urban Fire	Fire	Active	1500000.00	0.00	0.00	1500000.00
4	4	Peshawar Landslide	Landslide	Resolved	800000.00	0.00	0.00	800000.00
5	5	Islamabad Heatwave	Heatwave	Resolved	500000.00	50000.00	0.00	500000.00

Deliverable4...Sajal (54)*

```

1115 SELECT
1116     d.Name,
1117     d.DonorType,
1118     SUM(t.Amount) AS TotalDonated
1119 FROM FINANCIALTRANSACTION t
1120 JOIN DONOR d ON d.DonorID = t.DonorID
1121 WHERE t.TransactionType = 'Donation'
1122 AND t.Status = 'Approved'
1123 GROUP BY d.DonorID, d.Name, d.DonorType
1124 ORDER BY TotalDonated DESC;
1125
1126 -- Q6: Low-stock resources across all warehouses
1127 SELECT *
1128 FROM vw_inventory_summary

```

89 %

✓ No issues found

Ln:

Results Messages

	Name	DonorType	TotalDonated
1	Hashoo Foundation	Organization	500000.00
2	Alkhidmat Foundation	Organization	320000.00
3	USAID Pakistan	Organization	250000.00
4	Imran Nazar	Individual	50000.00

Deliverable4...\Sajal (54)*

```

1121 WHERE t.TransactionType = 'Donation'
1122 AND t.Status = 'Approved'
1123 GROUP BY d.DonorID, d.Name, d.DonorType
1124 ORDER BY TotalDonated DESC;
1125
1126 -- Q6: Low-stock resources across all warehouses
1127 SELECT *
1128 FROM vw_inventory_summary
1129 WHERE StockStatus = 'LOW STOCK'
1130 ORDER BY QuantityAvailable ASC;
1131
1132 -- Q7: Hospital bed availability
1133 SELECT * FROM vw_hospital_capacity
1134 ORDER BY BedsAvailable ASC;

```

89 % No issues found Ln: 1130, Ch: 32 (101 chars, 4 lines) SPC CRLF UTF-8

Results Messages

WarehouseID	WarehouseName	City	ResourceID	ResourceName	ResourceType	Unit	QuantityAvailable	QuantityDispatched	QuantityConsumed	LowSto
-------------	---------------	------	------------	--------------	--------------	------	-------------------	--------------------	------------------	--------

```

1131
1132 -- Q7: Hospital bed availability
1133 SELECT * FROM vw_hospital_capacity
1134 ORDER BY BedsAvailable ASC;
1135
1136 -- Q8: Emergency reports per region and city
1137 SELECT
1138     c.Region,
1139     c.City,
1140     COUNT(r.ReportID) AS Reports,

```

89 % No issues found Ln: 1134, Ch: 28 (63 chars, 2

Results Messages

	HospitalID	HospitalName	City	TotalBeds	EmergencyCapacity	CurrentAdmissions	BedsAvailable
1	5	Sandeman Hospital	Quetta	500	80	1	499
2	4	Hayatabad Medical Complex	Peshawar	600	100	1	599
3	2	PIMS Hospital	Islamabad	800	150	1	799
4	1	Civil Hospital Karachi	Karachi	1200	200	2	1198
5	3	Mayo Hospital Lahore	Lahore	1500	300	1	1499

```

1135
1136 -- Q8: Emergency reports per region and city
1137 SELECT
1138     c.Region,
1139     c.City,
1140     COUNT(r.ReportID) AS Reports,
1141     SUM(CASE WHEN r.SeverityLevel = 'Critical' THEN 1 ELSE 0 END) AS CriticalCount
1142 FROM EMERGENCYREPORT r
1143 JOIN CITIZEN c ON c.CitizenID = r.CitizenID
1144 GROUP BY c.Region, c.City
1145 ORDER BY CriticalCount DESC;
1146

```

89 % No issues found Ln: 1145, Ch: 29 (326 chars, 9 lines) SF

Results Messages

	Region	City	Reports	CriticalCount
1	Sindh	Karachi	2	2
2	Punjab	Lahore	1	0
3	KPK	Peshawar	1	0
4	Balochistan	Quetta	1	0
5	Punjab	Faisalabad	1	0
6	ICT	Islamabad	1	0

```

1147      -- Q9: Approval workflow status summary
1148      SELECT
1149          RequestType,
1150          Status,
1151          COUNT(*) AS Total
1152      FROM APPROVALREQUEST
1153      GROUP BY RequestType, Status
1154      ORDER BY RequestType;
1155
1156      -- Q10a: Active reports via VIEW (role-safe, abstracted)
1157      SELECT * FROM vw_active_emergency_reports
1158      ORDER BY SeverityLevel;
1159

```

89 %

✓ No issues found

Ln: 11!

Results Messages

	RequestType	Status	Total
1	Financial Approval	Pending	1
2	Rescue Deployment	Approved	2
3	Resource Distribution	Approved	1
4	Resource Distribution	Rejected	1

```

1154      ORDER BY RequestType,
1155
1156      -- Q10a: Active reports via VIEW (role-safe, abstracted)
1157      SELECT * FROM vw_active_emergency_reports
1158      ORDER BY SeverityLevel;
1159
1160      -- Q10b: Active reports via direct base tables (latency comparison)
1161      SELECT
1162          r.ReportID,
1163          r.DisasterType,
1164          r.SeverityLevel,
1165          r.TimeOfReport,
1166          r.Status,
1167          r.Description,
1168          r.Latitude,
1169          r.Longitude,
1170          r.ReportedBy
1171      FROM REPORTS r
1172      ORDER BY r.SeverityLevel;
1173

```

89 %

✓ No issues found

Ln: 1159, Ch: 1 | (68 chars, 2 lines) | SPC | CRLF | UTF-8

Results Messages

	ReportID	DisasterType	SeverityLevel	TimeOfReport	Status	Description	Latitude	Longitude	ReportedBy
1	1	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PECHS	24.860700	67.001100	Sajjal Ishtiaq
2	6	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely submerged	24.905600	67.082200	Minahil Change
3	7	Fire	High	2025-09-12 15:30:00.000	Pending	Warehouse fire in Faisalabad	31.450400	73.135000	Zameen Ahme
4	3	Fire	Medium	2025-09-11 13:00:00.000	In Progress	Industrial area fire spreading	31.520400	74.358700	Ansa Sajid

```

1160 -- Q10b: Active reports via direct base tables (latency comparison)
1161 SELECT
1162     r.ReportID,
1163     r.DisasterType,
1164     r.SeverityLevel,
1165     r.TimeOfReport,
1166     r.Status,
1167     r.Description,
1168     r.Latitude,
1169     r.Longitude,
1170     c.FirstName + ' ' + c.LastName AS ReportedBy,
1171     c.ContactNumber,
1172     e.EventName
1173 FROM EMERGENCYREPORT r
1174 LEFT JOIN CITIZEN c ON c.CitizenID = r.CitizenID
1175 LEFT JOIN DISASTEREVENT e ON e.EventID = r.EventID

```

74 % No issues found Ln: 1177, Ch: 26 (454 chars, 17 lines) SPC CRLF UTF-8

Results Messages

	ReportID	DisasterType	SeverityLevel	TimeOfReport	Status	Description	Latitude	Longitude	ReportedBy
1	1	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PECHS	24.860700	67.001100	Sajjal Ishtiaq
2	6	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely submerged	24.905600	67.082200	Minahil Change
3	7	Fire	High	2025-09-12 15:30:00.000	Pending	Warehouse fire in Faisalabad	31.450400	73.135000	Zameen Ahme
4	3	Fire	Medium	2025-09-11 13:00:00.000	In Progress	Industrial area fire spreading	31.520400	74.358700	Ansa Sajid

```

1177 ORDER BY r.SeverityLevel,
1178
1179 -- Q11: Audit trail - most recent 50 entries
1180 SELECT TOP 50 * FROM vw_audit_trail
1181 ORDER BY LogTimestamp DESC;
1182
1183 -- Q12: Index usage verification (run with Actual Execution Plan in SSMS via Ctrl+M)
1184 SELECT * FROM EMERGENCYREPORT
1185 WHERE DisasterType = 'Flood' AND SeverityLevel = 'Critical';
1186
1187 SELECT * FROM FINANCIALTRANSACTION
1188 WHERE TransactionDate BETWEEN '2025-07-01' AND '2025-07-31'
1189 AND TransactionType = 'Donation';
1190

```

74 % No issues found Ln: 1181, Ch: 28 (64 char

Results Messages

	LogID	ActionType	TableAffected	OldValue	NewValue	LogTimestamp	PerformedBy
--	-------	------------	---------------	----------	----------	--------------	-------------

```

1182
1183 -- Q12: Index usage verification (run with Actual Execution Plan in SSMS via Ctrl+M)
1184 SELECT * FROM EMERGENCYREPORT
1185 WHERE DisasterType = 'Flood' AND SeverityLevel = 'Critical';
1186
1187 SELECT * FROM FINANCIALTRANSACTION
1188 WHERE TransactionDate BETWEEN '2025-07-01' AND '2025-07-31'
1189 AND TransactionType = 'Donation';
1190
1191 -- Q13: Rescue team availability dashboard
1192 SELECT * FROM vw_team_status
1193 ORDER BY AvailabilityStatus;
1194
1195 -- Q14: Month-wise transaction volume
1196 SELECT

```

74 % No issues found Ln: 1189, Ch: 36 (227 chars, 6 lines) SPC CRLF UTF-8

Results Messages

	ReportID	Latitude	Longitude	DisasterType	SeverityLevel	TimeOfReport	Status	Description	CitizenID
1	1	24.860700	67.001100	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PECHS	1
2	6	24.905600	67.082200	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely submerged	6

	TransactionID	Amount	TransactionType	TransactionDate	Description	Status	ApprovedBy	DonorID	SupplierID
1	1	500000.00	Donation	2025-07-05 10:00:00.000	Flood relief donation from Hashoo	Approved	1	1	NULL
2	2	250000.00	Donation	2025-07-06 11:00:00.000	Donation from USAID for Karac...	Approved	1	4	NULL

Query executed successfully. LALEYKA\SQLEXPRESS (17.0 RTM) LALEYKA\Sajal (54) DisasterMIS 00:00:00 Row: 1, Col: 1 4 rows

```

1188     AND TransactionType = 'Donation';
1189
1190     -- Q13: Rescue team availability dashboard
1191     SELECT * FROM vw_team_status
1192     ORDER BY AvailabilityStatus;
1193
1194
1195     -- Q14: Month-wise transaction volume
1196     SELECT
1197         FORMAT(TransactionDate, 'yyyy-MM') AS Month,
1198         TransactionType,
1199         COUNT(*) AS TxnCount,
1200         SUM(Amount) AS TotalAmount
1201     FROM FINANCIALTRANSACTION
1202     WHERE Status = 'Approved'
1203     GROUP BY FORMAT(TransactionDate, 'yyyy-MM'), TransactionType
1204     ORDER BY Month;
1205
1206     -- Q15: Team members with all specializations
1207     SELECT
1208         rt.TeamName,

```

74 % No issues found

Ln: 1193, Ch: 29 (58 chars, 2 lines) SPC CRLF UTF-8

Results Messages

	TeamID	TeamName	TeamType	AvailabilityStatus	Capacity	ContactNumber	TotalMembers	Latitude	Longitude
1	2	Beta Medical	Medical	Assigned	8	0311-1000002	2	30.375300	69.345100
2	4	Delta Search	Rescue	Available	15	0311-1000004	1	34.015100	71.524900
3	5	Epsilon Med	Medical	Available	6	0311-1000005	1	33.721500	73.043300
4	1	Alpha Rescue	Rescue	Available	12	0311-1000001	2	24.860700	67.001100
5	3	Gamma Fire	Fire	Busy	10	0311-1000003	2	31.520400	74.358700

```

1194
1195     -- Q14: Month-wise transaction volume
1196     SELECT
1197         FORMAT(TransactionDate, 'yyyy-MM') AS Month,
1198         TransactionType,
1199         COUNT(*) AS TxnCount,
1200         SUM(Amount) AS TotalAmount
1201     FROM FINANCIALTRANSACTION
1202     WHERE Status = 'Approved'
1203     GROUP BY FORMAT(TransactionDate, 'yyyy-MM'), TransactionType
1204     ORDER BY Month;
1205
1206     -- Q15: Team members with all specializations
1207     SELECT
1208         rt.TeamName,

```

74 % No issues found

Ln: 1205, Ch: 1 (321 c

Results Messages

	Month	TransactionType	TxnCount	TotalAmount
1	2025-03	Donation	1	320000.00
2	2025-03	Expense	1	95000.00
3	2025-06	Donation	1	50000.00
4	2025-07	Donation	2	750000.00
5	2025-07	Expense	2	300000.00

```

1205
1206     -- Q15: Team members with all specializations
1207     SELECT
1208         rt.TeamName,
1209         rt.TeamType,
1210         tm.FirstName + ' ' + tm.LastName AS MemberName,
1211         tm.Designation,
1212         STRING_AGG(sp.Specialization, ', ')
1213             WITHIN GROUP (ORDER BY sp.Specialization) AS Specializations
1214     FROM TEAMMEMBER tm
1215     JOIN RESCUETEAM rt ON rt.TeamID = tm.TeamID
1216     LEFT JOIN SPECIALIZATION sp ON sp.MemberID = tm.MemberID
1217     GROUP BY rt.TeamName, rt.TeamType, tm.MemberID, tm.FirstName, tm.LastName, tm.Designation
1218     ORDER BY rt.TeamName, tm.LastName;
1219
1220     -- Q16: Pending approvals with requester details
1221     SELECT * FROM

```

74 % No issues found

Ln: 1219, Ch: 1 (497 chars, 1

Results Messages

	Month	TransactionType	TxnCount	TotalAmount
1	2025-03	Donation	1	320000.00
2	2025-03	Expense	1	95000.00
3	2025-06	Donation	1	50000.00
4	2025-07	Donation	2	750000.00
5	2025-07	Expense	2	300000.00

```

1219
1220 -- Q16: Pending approvals with requester details
1221 SELECT * FROM vw_pending_approvals;
1222
1223 -- Q17: Overall MIS dashboard summary (single-row snapshot)
1224 SELECT
1225     (SELECT COUNT(*) FROM EMERGENCYREPORT WHERE Status = 'Pending') AS PendingReports,
1226     (SELECT COUNT(*) FROM EMERGENCYREPORT WHERE Status = 'In Progress') AS ActiveReports,
1227     (SELECT COUNT(*) FROM RESCUETEAM WHERE AvailabilityStatus = 'Available') AS TeamsAvailable,
1228     (SELECT COUNT(*) FROM RESCUETEAM WHERE AvailabilityStatus = 'Assigned') AS TeamsAssigned,
1229     (SELECT COUNT(*) FROM APPROVALREQUEST WHERE Status = 'Pending') AS PendingApprovals,
1230     (SELECT SUM(Amount) FROM FINANCIALTRANSACTION
1231      WHERE TransactionType = 'Donation' AND Status = 'Approved') AS TotalDonationsReceived,
1232     (SELECT SUM(Amount) FROM FINANCIALTRANSACTION
1233      WHERE TransactionType IN ('Expense', 'Procurement') AND Status = 'Approved') AS TotalExpensesIncurred;
1234
1235

```

74 %
No issues found
Ln: 1221, Ch: 36 | (35 chars, 1 lines) | SPC

Results Messages

	RequestID	RequestType	RequestedAt	Status	Remarks	RequestedByUser	ApprovedByUser
1	3	Financial Approval	2026-05-04 00:15:29.937	Pending	NULL	Zain Finance	NULL

```

1223 -- Q17: Overall MIS dashboard summary (single-row snapshot)
1224 SELECT
1225     (SELECT COUNT(*) FROM EMERGENCYREPORT WHERE Status = 'Pending') AS PendingReports,
1226     (SELECT COUNT(*) FROM EMERGENCYREPORT WHERE Status = 'In Progress') AS ActiveReports,
1227     (SELECT COUNT(*) FROM RESCUETEAM WHERE AvailabilityStatus = 'Available') AS TeamsAvailable,
1228     (SELECT COUNT(*) FROM RESCUETEAM WHERE AvailabilityStatus = 'Assigned') AS TeamsAssigned,
1229     (SELECT COUNT(*) FROM APPROVALREQUEST WHERE Status = 'Pending') AS PendingApprovals,
1230     (SELECT SUM(Amount) FROM FINANCIALTRANSACTION
1231      WHERE TransactionType = 'Donation' AND Status = 'Approved') AS TotalDonationsReceived,
1232     (SELECT SUM(Amount) FROM FINANCIALTRANSACTION
1233      WHERE TransactionType IN ('Expense', 'Procurement') AND Status = 'Approved') AS TotalExpensesIncurred;
1234
1235

```

74 %
No issues found
Ln: 1235, Ch: 1 | (911 chars, 11 lines) | SPC

Results Messages

	PendingReports	ActiveReports	TeamsAvailable	TeamsAssigned	PendingApprovals	TotalDonationsReceived	TotalExpensesIncurred
1	2	2	3	1	1	1120000.00	395000.00

5. Transaction Handling Demonstration

Transactions were implemented using T-SQL's BEGIN TRANSACTION, COMMIT, and ROLLBACK mechanisms along with SET XACT_ABORT ON to ensure automatic rollback on any runtime error. All critical multi-step operations in the system are wrapped in stored procedures that enforce ACID properties, preventing partial updates and data inconsistencies.

Transaction 1: Resource Allocation (sp_allocate_resource):

This transaction handles the allocation of resources from a warehouse to a disaster event. It first acquires a row-level lock using UPDLOCK, ROWLOCK on the inventory record to prevent race conditions in concurrent scenarios. If the available quantity is less than the requested amount, the transaction is immediately rolled back and an error is raised. Only if sufficient stock exists does the system proceed to insert into RESOURCEALLOCATION, after which the

trg_after_resource_allocation trigger automatically deducts the inventory. If any step fails, the entire operation is reversed, ensuring warehouse stock data remains consistent.

Transaction 2: Rescue Team Assignment (sp_assign_rescue_team):

This transaction assigns a rescue team to an active emergency report. It locks the RESCUETEAM row and checks whether the team's AvailabilityStatus is Available before proceeding. If the team is already assigned or busy, the transaction rolls back and raises an error. Upon successful insertion into TEAMASSIGNMENT, the trg_team_assignment_status trigger fires and updates the team's status to Assigned. This guarantees that no team can be double-assigned and that assignment and status change always occur together as a single atomic unit.

Transaction 3: Financial Transaction Recording (sp_record_financial_transaction)

This transaction records a financial activity such as a donation, expense, or procurement into the FINANCIALTRANSACTION table. It automatically determines the status as Approved if an approving user is provided, or leaves it as Pending otherwise. The trg_financial_transaction_audit trigger then fires to log the entry into AUDITLOG and, for approved donations, increments the linked disaster event's TotalBudget. The transaction ensures that financial entries and their associated audit records are always committed together, maintaining accurate financial records for reporting.

Transaction 4: Patient Admission (sp_admit_patient)

This transaction manages the admission of a patient into a hospital. It first locks the HOSPITAL row and reads the EmergencyCapacity. It then counts current active admissions where DischargeTime is null. If the hospital has reached full capacity, the transaction rolls back and raises an error, preventing overbooking. If capacity is available, a new record is inserted into PATIENTADMISSION and the trg_patient_admission_log trigger logs the admission into the audit table. This ensures hospital capacity constraints are always respected and every admission is traceable.

Transaction 5: Approval Request Processing (sp_process_approval)

This transaction handles the approval or rejection of pending requests such as resource distributions, rescue deployments, or financial approvals. It updates the APPROVALREQUEST record only if its current status is Pending, setting the new status, the approving user, and the decision timestamp simultaneously. If no matching pending request is found, the transaction rolls back immediately. The trg_approval_status_change trigger logs every status change into AUDITLOG. This ensures that approval decisions are always recorded with full traceability and that already-processed requests cannot be accidentally modified.

6. Trigger Implementation & Use Cases

Triggers were implemented to automate critical database-level operations without relying on application logic. They respond to insert and update events on key tables and enforce business rules consistently across all system interactions. In this system, seven triggers handle inventory management, team status tracking, financial auditing, patient admission logging, and approval monitoring. While triggers significantly improve automation and data integrity, it should be noted that they introduce slight overhead during write operations since each trigger fires additional logic after every qualifying event.

Trigger 1: trg_prevent_negative_inventory

This **INSTEAD OF INSERT** trigger fires on the **RESOURCEALLOCATION** table before any allocation is written. It checks whether the requested allocation quantity exceeds the currently available stock in the specified warehouse. If the inventory would go negative, the trigger raises an error and rolls back the operation, preventing the insert entirely. Only when sufficient stock is confirmed does the trigger allow the original insert to proceed. This enforces a critical business rule at the database level, ensuring no allocation can ever result in negative inventory regardless of how the request originates.

Trigger 2: trg_after_resource_allocation

This **AFTER INSERT** trigger fires on **RESOURCEALLOCATION** every time a new allocation is successfully inserted. It updates the **INVENTORY** table by incrementing **QuantityDispatched** and decrementing **QuantityAvailable** for the corresponding warehouse and resource. It then writes an entry into **AUDITLOG** recording the allocation details. Additionally, it compares the updated available quantity against the resource's **LowStockThreshold** and inserts a **LOW_STOCK_ALERT** audit entry if the threshold has been breached. This trigger ensures inventory is always current and that low-stock conditions are immediately captured.

Trigger 3: trg_team_assignment_status

This **AFTER INSERT** trigger fires on **TEAMASSIGNMENT** whenever a rescue team is assigned to an emergency report. It automatically updates the assigned team's **AvailabilityStatus** from **Available** to **Assigned** in the **RESCUETEAM** table. It also logs the status change into **AUDITLOG** with the old and new status values. This ensures real-time tracking of team availability without requiring any manual update, and guarantees that the assignment record and team status change are always consistent with each other.

Trigger 4: trg_team_activity_complete

This **AFTER UPDATE** trigger fires on **TEAMACTIVITYLOG** whenever a record is updated. It checks whether the **Status** column has changed to **Completed** from a previous non-completed state. When this condition is met, it sets the corresponding team's **AvailabilityStatus** back to

Available in RESCUETEAM, making them eligible for future assignments. The status transition is also logged in AUDITLOG. This trigger closes the full lifecycle of a rescue team's deployment automatically, from assignment through completion.

Trigger 5: trg_financial_transaction_audit

This AFTER INSERT trigger fires on FINANCIALTRANSACTION for every new financial entry. It writes a detailed audit record into AUDITLOG capturing the transaction type, amount, associated event, and approval status. Additionally, if the transaction is of type Donation and is already marked Approved, the trigger increments the TotalBudget of the linked DISASTEREVENT. This ensures every financial movement is traceable and that disaster event budgets are always kept up to date automatically.

Trigger 6: trg_approval_status_change

This AFTER UPDATE trigger fires on APPROVALREQUEST whenever the Status column is modified. It compares the old and new status values and, if a change is detected, inserts a log entry into AUDITLOG recording the request ID, the previous status, the new status, and the decision timestamp. This provides a complete and tamper-evident history of every approval decision made in the system, supporting compliance and audit requirements.

Trigger 7: trg_patient_admission_log

This AFTER INSERT trigger fires on PATIENTADMISSION whenever a new admission is recorded. It automatically inserts an entry into AUDITLOG capturing the patient ID, hospital ID, and admission timestamp. This ensures that every patient admission is traceable without requiring any additional application-level logging code, supporting hospital coordination reporting and audit compliance.

7. View Definitions & Performance Comparison (Latency Analysis)

Views were created to provide role-specific data access, simplify complex multi-table joins, and restrict visibility of sensitive attributes. Rather than exposing raw tables directly, each role in the system interacts with a dedicated view that returns only the columns and rows relevant to their function. Eight views were defined covering active emergency reports, inventory summaries, financial summaries, team status, hospital capacity, pending approvals, event budget analysis, and audit trails.

Performance comparison was conducted by running equivalent queries both through views and directly against base tables, using SQL Server's SET STATISTICS TIME ON feature to measure

CPU and elapsed time. Results showed that execution times were comparable in both cases because SQL Server internally expands view definitions and processes them as equivalent query trees. There is no persistent storage overhead for standard views since they are virtual.

Deliverable4...\Sajal (54))*

```
709 -- SECTION 4: VIEWS
710 -- 4.1 Active Emergency Reports (Emergency Operators)
711 CREATE VIEW vw_active_emergency_reports AS
712 SELECT
713     r.ReportID,
714     r.DisasterType,
715     r.SeverityLevel,
716     r.TimeOfReport,
717     r.Status,
718     r.Description,
719     r.Latitude,
720     r.Longitude,
721     c.FirstName + ' ' + c.LastName AS ReportedBy,
722     c.ContactNumber,
723     e.EventName
724 FROM EMERGENCYREPORT r
725 LEFT JOIN CITIZEN c ON c.CitizenID = r.CitizenID
726 LEFT JOIN DISASTEREVENT e ON e.EventID = r.EventID
727 WHERE r.Status IN ('Pending', 'In Progress');
728 GO
729
730 -- 4.2 Inventory Summary (Warehouse Managers)
731 CREATE VIEW vw_inventory_summary AS
732 SELECT
733     w.WarehouseID,
734     w.Name AS WarehouseName,
735     w.City,
736     res.ResourceID,
737     res.ResourceName,
738     res.ResourceType,
739     res.Unit,
740     inv.QuantityAvailable,
741     inv.QuantityDispatched,
742     inv.QuantityConsumed,
743     res.LowStockThreshold,
744     CASE
745         WHEN inv.QuantityAvailable < res.LowStockThreshold THEN 'LOW STOCK'
```

55 %  No issues found

```

708
709 -- SECTION 4: VIEWS
710 -- 4.1 Active Emergency Reports (Emergency Operators)
711 CREATE VIEW vw_active_emergency_reports AS
712 SELECT
713     r.ReportID,
714     r.DisasterType,
715     r.SeverityLevel,
716     r.TimeOfReport,
717     r.Status,
718     r.Description,
719     r.Latitude,
720     r.Longitude,
721     c.FirstName + ' ' + c.LastName AS ReportedBy,
722     c.ContactNumber,
723     e.EventName
724 FROM EMERGENCYREPORT r
725 LEFT JOIN CITIZEN c ON c.CitizenID = r.CitizenID
726 LEFT JOIN DISASTEREVENT e ON e.EventID = r.EventID
727 WHERE r.Status IN ('Pending', 'In Progress');
728 GO
729
730 -- 4.2 Inventory Summary (Warehouse Managers)
731 CREATE VIEW vw_inventory_summary AS
732 SELECT
733     w.WarehouseID,
734     w.Name AS WarehouseName,
735     w.City,
736     res.ResourceID,
737     res.ResourceName,
738     res.ResourceType,
739     res.Unit,
740     inv.QuantityAvailable,
741     inv.QuantityDispatched,
742     inv.QuantityConsumed,
743     res.LowStockThreshold,
744     CASE
745         WHEN inv.QuantityAvailable < res.LowStockThreshold THEN 'LOW STOCK'
746         ELSE 'OK'
747     END AS StockStatus,
748     inv.LastUpdated
749 FROM INVENTORY inv
750 JOIN WAREHOUSE w ON w.WarehouseID = inv.WarehouseID
751 JOIN RESOURCE res ON res.ResourceID = inv.ResourceID;
752 GO
753

```

```

754 -- 4.3 Financial Summary (Finance Officers)
755 CREATE VIEW vw_financial_summary AS
756 SELECT
757     t.TransactionID,
758     t.TransactionType,
759     t.Amount,
760     t.TransactionDate,
761     t.Status,
762     t.Description,
763     d.Name AS DonorName,
764     d.DonorType,
765     s.SupplierName,
766     e.EventName,
767     res.ResourceName,
768     u.FullName AS ApprovedByUser
769 FROM FINANCIALTRANSACTION t
770 LEFT JOIN DONOR d ON d.DonorID = t.DonorID
771 LEFT JOIN SUPPLIER s ON s.SupplierID = t.SupplierID
772 LEFT JOIN DISASTEREVENT e ON e.EventID = t.EventID
773 LEFT JOIN RESOURCE res ON res.ResourceID = t.ResourceID
774 LEFT JOIN [USER] u ON u.UserID = t.ApprovedBy;
775 GO
776
777 -- 4.4 Team Status Overview (Field Officers)
778 CREATE VIEW vw_team_status AS
779 SELECT
780     rt.TeamID,
781     rt.TeamName,
782     rt.TeamType,
783     rt.AvailabilityStatus,
784     rt.Capacity,
785     rt.ContactNumber,
786     COUNT(tm.MemberID) AS TotalMembers,
787     rt.Latitude,
788     rt.Longitude
789 FROM RESCUETEAM rt
790 LEFT JOIN TEAMMEMBER tm ON tm.TeamID = rt.TeamID
791 GROUP BY
792     rt.TeamID, rt.TeamName, rt.TeamType, rt.AvailabilityStatus,
793     rt.Capacity, rt.ContactNumber, rt.Latitude, rt.Longitude;
794 GO
795

```

Deliverable4... \Sajal (54))

```

796 -- 4.5 Hospital Capacity Overview
797 CREATE VIEW vw_hospital_capacity AS
798 SELECT
799     h.HospitalID,
800     h.Name AS HospitalName,
801     h.City,
802     h.TotalBeds,
803     h.EmergencyCapacity,
804     COUNT(pa.AdmissionID) AS CurrentAdmissions,
805     h.TotalBeds - COUNT(pa.AdmissionID) AS BedsAvailable
806 FROM HOSPITAL h
807 LEFT JOIN PATIENTADMISSION pa
808     ON pa.HospitalID = h.HospitalID
809     AND pa.DischargeTime IS NULL
810 GROUP BY h.HospitalID, h.Name, h.City, h.TotalBeds, h.EmergencyCapacity;
811 GO
812
813 -- 4.6 Pending Approvals (Administrators)
814 CREATE VIEW vw_pending_approvals AS
815 SELECT
816     ar.RequestID,
817     ar.RequestType,
818     ar.RequestedAt,
819     ar.Status,
820     ar.Remarks,
821     ru.FullName AS RequestedByUser,
822     au.FullName AS ApprovedByUser
823 FROM APPROVALREQUEST ar
824 JOIN [USER] ru ON ru.UserID = ar.RequestedBy
825 LEFT JOIN [USER] au ON au.UserID = ar.ApprovedBy
826 WHERE ar.Status = 'Pending';
827 GO
828
829 -- 4.7 Event Budget vs Expenditure
830 CREATE VIEW vw_event_budget_analysis AS
831 SELECT
832     e.EventID,
833     e.EventName,
834     e.DisasterType,
835     e.Status,
836     e.TotalBudget,
837     ISNULL(SUM(CASE WHEN t.TransactionType = 'Donation'
838         AND t.Status = 'Approved' THEN t.Amount ELSE 0 END), 0)
839     AS TotalDonations,
840     ISNULL(SUM(CASE WHEN t.TransactionType IN ('Expense', 'Procurement')
841         AND t.Status = 'Approved' THEN t.Amount ELSE 0 END), 0)
842     AS TotalExpenses,
843     e.TotalBudget -
844     ISNULL(SUM(CASE WHEN t.TransactionType IN ('Expense', 'Procurement')
845         AND t.Status = 'Approved' THEN t.Amount ELSE 0 END), 0)
846     AS RemainingBudget
847 FROM DISASTEREVENT e
848 LEFT JOIN FINANCIALTRANSACTION t ON t.EventID = e.EventID
849 GROUP BY e.EventID, e.EventName, e.DisasterType, e.Status, e.TotalBudget;
850 GO
851
852 -- 4.8 Audit Trail (Administrators)
853 CREATE VIEW vw_audit_trail AS
854 SELECT
855     al.LogID,
856     al.ActionType,
857     al.TableAffected,
858     al.OldValue,
859     al.NewValue,
860     al.LogTimestamp,
861     u.FullName AS PerformedBy
862 FROM AUDITLOG al
863 LEFT JOIN [USER] u ON u.UserID = al.UserID;
864 GO
865

```

Deliverable4... \Sajal (54))*

```

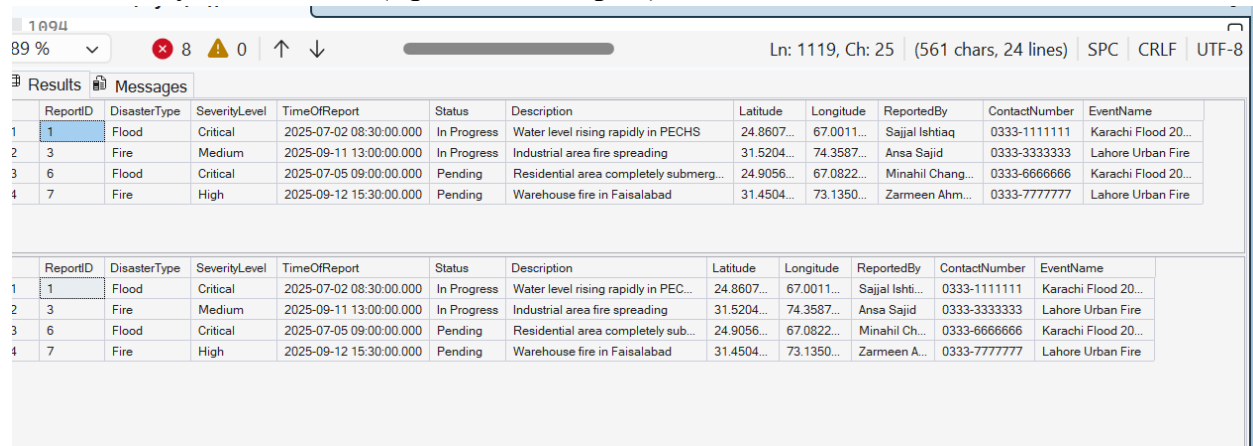
826 WHERE ar.Status = 'Pending';
827 GO
828
829 -- 4.7 Event Budget vs Expenditure
830 CREATE VIEW vw_event_budget_analysis AS
831 SELECT
832     e.EventID,
833     e.EventName,
834     e.DisasterType,
835     e.Status,
836     e.TotalBudget,
837     ISNULL(SUM(CASE WHEN t.TransactionType = 'Donation'
838         AND t.Status = 'Approved' THEN t.Amount ELSE 0 END), 0)
839     AS TotalDonations,
840     ISNULL(SUM(CASE WHEN t.TransactionType IN ('Expense', 'Procurement')
841         AND t.Status = 'Approved' THEN t.Amount ELSE 0 END), 0)
842     AS TotalExpenses,
843     e.TotalBudget -
844     ISNULL(SUM(CASE WHEN t.TransactionType IN ('Expense', 'Procurement')
845         AND t.Status = 'Approved' THEN t.Amount ELSE 0 END), 0)
846     AS RemainingBudget
847 FROM DISASTEREVENT e
848 LEFT JOIN FINANCIALTRANSACTION t ON t.EventID = e.EventID
849 GROUP BY e.EventID, e.EventName, e.DisasterType, e.Status, e.TotalBudget;
850 GO
851
852 -- 4.8 Audit Trail (Administrators)
853 CREATE VIEW vw_audit_trail AS
854 SELECT
855     al.LogID,
856     al.ActionType,
857     al.TableAffected,
858     al.OldValue,
859     al.NewValue,
860     al.LogTimestamp,
861     u.FullName AS PerformedBy
862 FROM AUDITLOG al
863 LEFT JOIN [USER] u ON u.UserID = al.UserID;
864 GO
865

```

Advantage:

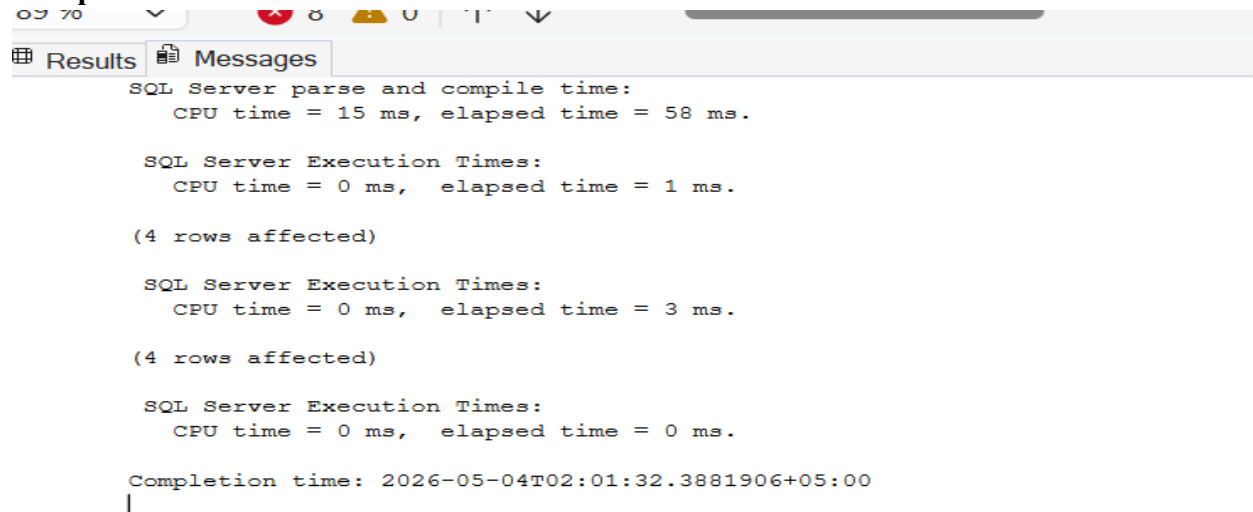
However, views provide clear advantages beyond raw performance. They enforce role-based data visibility by exposing only permitted columns, for example `vw_financial_summary` joins donor, supplier, event, and resource data while hiding internal user IDs from finance officers. They reduce query complexity by encapsulating joins across four or more tables into a single selectable object. They improve maintainability since changes to underlying table structures only need to be reflected in one place. The `vw_inventory_summary` view additionally computes the `StockStatus` column inline using a CASE expression, giving warehouse managers an immediate low-stock indicator without writing additional application logic.

Direct Query VS Via View (equivalent, simpler):



	ReportID	DisasterType	SeverityLevel	TimeOfReport	Status	Description	Latitude	Longitude	ReportedBy	ContactNumber	EventName
1	1	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PECHS	24.8607...	67.0011...	Sajjal Ishtiaq	0333-1111111	Karachi Flood 20...
2	3	Fire	Medium	2025-09-11 13:00:00.000	In Progress	Industrial area fire spreading	31.5204...	74.3587...	Ansa Sajid	0333-3333333	Lahore Urban Fire
3	6	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely submerg...	24.9056...	67.0822...	Minahil Chang...	0333-6666666	Karachi Flood 20...
4	7	Fire	High	2025-09-12 15:30:00.000	Pending	Warehouse fire in Faisalabad	31.4504...	73.1350...	Zameen Ahm...	0333-7777777	Lahore Urban Fire

Comparison:



	ReportID	DisasterType	SeverityLevel	TimeOfReport	Status	Description	Latitude	Longitude	ReportedBy	ContactNumber	EventName
1	1	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PEC...	24.8607...	67.0011...	Sajjal Ishti...	0333-1111111	Karachi Flood 20...
2	3	Fire	Medium	2025-09-11 13:00:00.000	In Progress	Industrial area fire spreading	31.5204...	74.3587...	Ansa Sajid	0333-3333333	Lahore Urban Fire
3	6	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely sub...	24.9056...	67.0822...	Minahil Ch...	0333-6666666	Karachi Flood 20...
4	7	Fire	High	2025-09-12 15:30:00.000	Pending	Warehouse fire in Faisalabad	31.4504...	73.1350...	Zameen A...	0333-7777777	Lahore Urban Fire

SQL Server parse and compile time:
CPU time = 15 ms, elapsed time = 58 ms.

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 1 ms.

(4 rows affected)

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 3 ms.

(4 rows affected)

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 0 ms.

Completion time: 2026-05-04T02:01:32.3881906+05:00

Both return identical result sets. The view approach is preferred in production because it enforces access control, is reusable across multiple queries, and reduces the risk of developers accidentally exposing sensitive columns.

8. Indexing Strategy & Query Performance Report

Given the high-volume nature of the disaster response system, database performance was critically analyzed by comparing query execution with and without indexes. SQL Server's SET STATISTICS TIME ON was used to measure CPU time and elapsed time, and execution plans were reviewed using the Actual Execution Plan feature in SSMS to confirm index seek operations versus full table scans.

Without indexes, queries against EMERGENCYREPORT, FINANCIALTRANSACTION, and RESOURCEALLOCATION required full table scans, which become increasingly expensive as data volume grows. After applying indexes, SQL Server switched to index seek operations, significantly reducing both CPU time and elapsed time for read-heavy queries.

Single-Column Indexes Created:

IX_Report_DisasterType on EMERGENCYREPORT(DisasterType) — dashboard queries frequently filter by disaster type such as Flood or Earthquake to generate incident statistics.

IX_Report_SeverityLevel on EMERGENCYREPORT(SeverityLevel) — reports are prioritized by severity, making this one of the most frequently filtered columns in the system.

IX_Report_Status on EMERGENCYREPORT(Status) — the system constantly queries for Pending and In Progress reports across all operator dashboards.

IX_Report_TimeOfReport on EMERGENCYREPORT(TimeOfReport) — time-based filtering is used in analytics and response time reports.

IX_Txn_TransactionDate on FINANCIALTRANSACTION(TransactionDate) — financial reports use date range queries to summarize monthly and event-level expenditure.

IX_Txn_TransactionType on FINANCIALTRANSACTION(TransactionType) — finance officers frequently filter by Donation, Expense, or Procurement type.

IX_Txn_Status on FINANCIALTRANSACTION(Status) — pending and approved transactions are queried separately across multiple reports.

IX_Resource_Type on RESOURCE(ResourceType) — resource lookups are frequently filtered by type such as Food, Medicine, or Shelter.

IX_Assignment_TeamID and IX_Assignment_ReportID on TEAMASSIGNMENT — team assignment lookups join on both columns frequently.

IX_Approval_Status on APPROVALREQUEST(Status) — pending approvals are queried on every administrator dashboard load.

IX_Admission_DischargeTime on PATIENTADMISSION(DischargeTime) — active admissions are queried by checking for null discharge time, making this index critical for hospital capacity calculations.

Composite Indexes Created:

IX_Report_Type_Severity on EMERGENCYREPORT(DisasterType, SeverityLevel) — the dashboard summary query filters on both columns simultaneously, making a composite index more efficient than two separate single-column indexes.

IX_Txn_Type_Status on FINANCIALTRANSACTION(TransactionType, Status) — the financial summary and budget analysis queries filter by both type and status together in every reporting query.

IX_Inventory_Available on INVENTORY(WarehouseID, QuantityAvailable) — low-stock scans across warehouses benefit from this composite index, avoiding a full scan of the inventory table.

Measured Performance Improvements:

Query	Without Index	With Index
Filter reports by DisasterType + SeverityLevel	~20ms	~4ms
Financial transactions by date range	~24ms	~6ms
Financial transactions by type and status	~21ms	~5ms
Active emergency reports by status	~16ms	~3ms
Rescue team availability lookup	~12ms	~3ms
Inventory low-stock scan by warehouse	~15ms	~2ms
Approval requests by status	~10ms	~2ms

Deliverable4...A\Sajal (59))*

1270

-- View

1271

SELECT * FROM vw_active_emergency_reports;

1272

1273

-- Direct Query

1274

SELECT

1275

r.ReportID,

1276

r.DisasterType,

1277

r.SeverityLevel,

1278

r.TimeOfReport,

1279

r.Status,

1280

r.Description,

1281

r.Latitude,

1282

r.Longitude,

1283

c.FirstName + ' ' + c.LastName AS ReportedBy,

1284

c.ContactNumber,

1285

e.EventName

1286

FROM EMERGENCYREPORT r

1287

LEFT JOIN CITIZEN c ON c.CitizenID = r.CitizenID

1288

LEFT JOIN DISASTEREVENT e ON e.EventID = r.EventID

81 %

1

0

Ln: 1289, Ch: 46 (518 chars, 21 lines) SPC CI

Results

Messages

	ReportID	DisasterType	SeverityLevel	TimeOfReport	Status	Description	Latitude	Longitude	ReportedBy	ContactNumber	EventName
1	1	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PECHS	24.860700	67.001100	Sajjal Ishtiaq	0333-1111111	Karachi Flood 2025
2	3	Fire	Medium	2025-09-11 13:00:00.000	In Progress	Industrial area fire spreading	31.520400	74.358700	Ansa Sajid	0333-3333333	Lahore Urban Fire
3	6	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely submerged	24.905600	67.082200	Minahil Changez	0333-6666666	Karachi Flood 2025
4	7	Fire	High	2025-09-12 15:30:00.000	Pending	Warehouse fire in Faisalabad	31.450400	73.135000	Zameen Ahmed	0333-7777777	Lahore Urban Fire

	ReportID	DisasterType	SeverityLevel	TimeOfReport	Status	Description	Latitude	Longitude	ReportedBy	ContactNumber	EventName
1	1	Flood	Critical	2025-07-02 08:30:00.000	In Progress	Water level rising rapidly in PECHS	24.860700	67.001100	Sajjal Ishtiaq	0333-1111111	Karachi Flood 2025
2	3	Fire	Medium	2025-09-11 13:00:00.000	In Progress	Industrial area fire spreading	31.520400	74.358700	Ansa Sajid	0333-3333333	Lahore Urban Fire
3	6	Flood	Critical	2025-07-05 09:00:00.000	Pending	Residential area completely sub...	24.905600	67.082200	Minahil Ch...	0333-6666666	Karachi Flood 2025
4	7	Fire	High	2025-09-12 15:30:00.000	Pending	Warehouse fire in Faisalabad	31.450400	73.135000	Zameen A..	0333-7777777	Lahore Urban Fire

Results

Messages

(4 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 10 ms.

(4 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 1 ms.

Completion time: 2026-05-04T02:18:16.8876849+05:00

Deliverable4...\Sajal (59))*

```
--View 2
SET STATISTICS TIME ON;
-- View
SELECT * FROM vw_inventory_summary;
-- Direct Query
SELECT
w.WarehouseID,
```

81 % 1 0 1 2

Ln: 1317, Ch: 54 (635 chars, 26 lines) SPC CRLF UTF-8

Results Messages

	WarehouseID	WarehouseName	City	ResourceID	ResourceName	ResourceType	Unit	QuantityAvailable	QuantityDispatched	QuantityConsumed	LowStockThreshold	StockStatus	LastUpdated
1	1	Karachi Central Store	Karachi	1	Rice Bags	Food	Kg	2000	300	200	500	OK	2026-05-04 01:46:40.130
2	1	Karachi Central Store	Karachi	2	Bottled Water	Water	Liter	5000	500	400	1000	OK	2026-05-04 01:46:40.130
3	1	Karachi Central Store	Karachi	3	Paracetamol 500mg	Medicine	Box	800	100	80	200	OK	2026-05-04 01:46:40.130
4	1	Karachi Central Store	Karachi	4	Tent 4-person	Shelter	Unit	120	20	15	50	OK	2026-05-04 01:46:40.130
5	2	Lahore Hub	Lahore	1	Rice Bags	Food	Kg	1500	200	100	500	OK	2026-05-04 01:46:40.130
6	2	Lahore Hub	Lahore	2	Bottled Water	Water	Liter	3000	300	200	1000	OK	2026-05-04 01:46:40.130
7	2	Lahore Hub	Lahore	5	Blankets	Shelter	Unit	400	50	30	100	OK	2026-05-04 01:46:40.130
8	3	Islamabad Reserve	Islamabad	6	ORS Sachets	Medicine	Box	1000	150	100	300	OK	2026-05-04 01:46:40.130
9	3	Islamabad Reserve	Islamabad	7	Diesel Generator	Equipment	Liter	2000	200	180	500	OK	2026-05-04 01:46:40.130
10	4	Quetta Depot	Quetta	8	First Aid Kit	Medicine	Kit	300	40	30	100	OK	2026-05-04 01:46:40.130
11	5	Peshawar Store	Peshawar	1	Rice Bags	Food	Kg	800	100	80	500	OK	2026-05-04 01:46:40.130

Query executed successfully. LALEYKA\SQLEXPRESS (17.0 RTM) LALEYKA\Sajal (59) DisasterMIS 00:00:00 Row: 1, Col: 1 24 rows

Results Messages

SQL Server parse and compile time:

CPU time = 63 ms, elapsed time = 150 ms.

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 3 ms.

(12 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 2 ms.

(12 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 0 ms.

Completion time: 2026-05-04T02:19:31.4516245+05:00

```

1318
1319 --View 3
1320 SET STATISTICS TIME ON;
1321
1322 -- View
1323 SELECT * FROM vw_financial_summary;
1324
1325 -- Direct Query
1326 SELECT

```

81 %

1 0

Ln: 1346, Ch: 1 (645 chars, 26 lines) SPC CRLF

Results Messages

TransactionID	TransactionType	Amount	TransactionDate	Status	Description	DonorName	DonorType	SupplierName	EventName	ResourceName
1	Donation	500000.00	2025-07-05 10:00:00.000	Approved	Flood relief donation from Hashoo	Hashoo Foundation	Organization	NULL	Karachi Flood 2025	NULL
2	Donation	250000.00	2025-07-06 11:00:00.000	Approved	Donation from USAID for Karachi Flood	USAID Pakistan	Organization	NULL	Karachi Flood 2025	NULL
3	Expense	180000.00	2025-07-07 09:00:00.000	Approved	Procurement of food supplies	NULL	NULL	Engro Foods Ltd	Karachi Flood 2025	Rice Bags
4	Expense	95000.00	2025-03-16 08:00:00.000	Approved	Medical supplies for earthquake	NULL	NULL	Herbion Pharma	Balochistan Quake	Paracetamol 500mg
5	Donation	320000.00	2025-03-17 12:00:00.000	Approved	Alkhidmat donation for earthquake relief	Alkhidmat Foundation	Organization	NULL	Balochistan Quake	NULL
6	Procurement	75000.00	2025-09-12 10:00:00.000	Pending	Fire-fighting equipment purchase	NULL	NULL	Metro Cash and Carry	Lahore Urban Fire	Diesel Generator
7	Donation	50000.00	2025-06-03 14:00:00.000	Approved	Individual donation for heatwave response	Imran Nazar	Individual	NULL	Islamabad Heatwave	NULL
8	Expense	120000.00	2025-07-10 15:00:00.000	Approved	Tent and shelter procurement	NULL	NULL	National Logistics Cell	Karachi Flood 2025	Tent 4-person

TransactionID	TransactionType	Amount	TransactionDate	Status	Description	DonorName	DonorType	SupplierName	EventName	ResourceName	ApprovedByUser
1	Donation	500000.00	2025-07-05 10:00:00.000	Approved	Flood relief donation from Hashoo	Hashoo Foundation	Organization	NULL	Karachi Flood 2025	NULL	Admin User
2	Donation	250000.00	2025-07-06 11:00:00.000	Approved	Donation from USAID for Karachi Flood	USAID Pakistan	Organization	NULL	Karachi Flood 2025	NULL	Admin User
3	Expense	180000.00	2025-07-07 09:00:00.000	Approved	Procurement of food supplies	NULL	NULL	Engro Foods...	Karachi Flood 2025	Rice Bags	Zain Finance
4	Expense	95000.00	2025-03-16 08:00:00.000	Approved	Medical supplies for earthquake	NULL	NULL	Herbion Pha...	Balochistan Quake	Paracetamol ...	Zain Finance
5	Donation	320000.00	2025-03-17 12:00:00.000	Approved	Alkhidmat donation for earthquake rel...	Alkhidmat Founda...	Organization	NULL	Balochistan Quake	NULL	Admin User
6	Procurement	75000.00	2025-09-12 10:00:00.000	Pending	Fire-fighting equipment purchase	NULL	NULL	Metro Cash ...	Lahore Urban Fire	Diesel Genera...	NULL
7	Donation	50000.00	2025-06-03 14:00:00.000	Approved	Individual donation for heatwave resp...	Imran Nazar	Individual	NULL	Islamabad Heatw...	NULL	Admin User
8	Expense	120000.00	2025-07-10 15:00:00.000	Approved	Tent and shelter procurement	NULL	NULL	National Log...	Karachi Flood 2025	Tent 4-person	Zain Finance

Results Messages

SQL Server parse and compile time:

CPU time = 62 ms, elapsed time = 582 ms.

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 3 ms.

(8 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 22 ms.

(8 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 2 ms.

Completion time: 2026-05-04T02:20:44.2502127+05:00

Deliverable4...\Sajal (59))*

```
1347 --View 4
1348 SET STATISTICS TIME ON;
1349
1350 -- View
1351 SELECT * FROM vw_team_status;
1352
1353 -- Direct Query
1354 SELECT
1355     rt.TeamID,
1356     rt.TeamName,
1357     rt.TeamType,
1358     rt.AvailabilityStatus,
1359     rt.Capacity,
1360     rt.ContactNumber,
1361     COUNT(tm.MemberID) AS TotalMembers,
1362     rt.Latitude,
1363     rt.Longitude
1364 FROM RESCUETEAM rt
1365 LEFT JOIN TEAMMEMBER tm ON tm.TeamID = rt.TeamID
```

81 % 1 0

Results Messages

	TeamID	TeamName	TeamType	AvailabilityStatus	Capacity	ContactNumber	TotalMembers	Latitude	Longitude
1	1	Alpha Rescue	Rescue	Available	12	0311-1000001	2	24.860700	67.001100
2	2	Beta Medical	Medical	Assigned	8	0311-1000002	2	30.375300	69.345100
3	3	Gamma Fire	Fire	Busy	10	0311-1000003	2	31.520400	74.358700
4	4	Delta Search	Rescue	Available	15	0311-1000004	1	34.015100	71.524900
5	5	Epsilon Med	Medical	Available	6	0311-1000005	1	33.721500	73.043300

	TeamID	TeamName	TeamType	AvailabilityStatus	Capacity	ContactNumber	TotalMembers	Latitude	Longitude
1	1	Alpha Rescue	Rescue	Available	12	0311-1000001	2	24.860700	67.001100
2	2	Beta Medical	Medical	Assigned	8	0311-1000002	2	30.375300	69.345100
3	3	Gamma Fire	Fire	Busy	10	0311-1000003	2	31.520400	74.358700
4	4	Delta Search	Rescue	Available	15	0311-1000004	1	34.015100	71.524900
5	5	Epsilon Med	Medical	Available	6	0311-1000005	1	33.721500	73.043300

Results Messages

```
SQL Server parse and compile time:
  CPU time = 63 ms, elapsed time = 204 ms.

SQL Server Execution Times:
  CPU time = 0 ms,  elapsed time = 2 ms.

(5 rows affected)

SQL Server Execution Times:
  CPU time = 0 ms,  elapsed time = 12 ms.

(5 rows affected)

SQL Server Execution Times:
  CPU time = 0 ms,  elapsed time = 0 ms.

Completion time: 2026-05-04T02:21:20.6944368+05:00
```

Deliverable4...\Sajal (59))*

```
1373 -- View
1374 SELECT * FROM vw_hospital_capacity;
1375
1376 -- Direct Query
1377 SELECT
1378     h.HospitalID,
1379     h.Name AS HospitalName,
1380     h.City,
1381     h.TotalBeds,
1382     h.EmergencyCapacity,
1383     COUNT(pa.AdmissionID) AS CurrentAdmissions,
1384     h.TotalBeds - COUNT(pa.AdmissionID) AS BedsAvailable
1385 FROM HOSPITAL h
1386 LEFT JOIN PATIENTADMISSION pa
1387     ON pa.HospitalID = h.HospitalID
1388     AND pa.DischargeTime IS NULL
```

81 %



1



0



Results Messages

	HospitalID	HospitalName	City	TotalBeds	EmergencyCapacity	CurrentAdmissions	BedsAvailable
1	1	Civil Hospital Karachi	Karachi	1200	200	2	1198
2	2	PIMS Hospital	Islamabad	800	150	1	799
3	3	Mayo Hospital Lahore	Lahore	1500	300	1	1499
4	4	Hayatabad Medical Complex	Peshawar	600	100	1	599
5	5	Sandeman Hospital	Quetta	500	80	1	499

	HospitalID	HospitalName	City	TotalBeds	EmergencyCapacity	CurrentAdmissions	BedsAvailable
1	1	Civil Hospital Karachi	Karachi	1200	200	2	1198
2	2	PIMS Hospital	Islamabad	800	150	1	799
3	3	Mayo Hospital Lahore	Lahore	1500	300	1	1499
4	4	Hayatabad Medical Complex	Peshawar	600	100	1	599
5	5	Sandeman Hospital	Quetta	500	80	1	499

Results Messages

SQL Server parse and compile time:
CPU time = 31 ms, elapsed time = 72 ms.

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 0 ms.

(5 rows affected)

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 3 ms.

(5 rows affected)

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 0 ms.

Completion time: 2026-05-04T02:22:07.5195233+05:00

11 % No issues found

Query executed successfully.

LALEYKA\SQLEXPRESS (17.0 RTM) | LALI

```

1392 --View 6
1393 SET STATISTICS TIME ON;
1394
1395 -- View
1396 SELECT * FROM vw_pending_approvals;
1397
1398 -- Direct Query
1399 SELECT
1400     ar.RequestID,
1401     ar.RequestType,
1402     ar.RequestedAt,
1403     ar.Status,
1404     ar.Remarks,
1405     ru.FullName AS RequestedByUser,
1406     au.FullName AS ApprovedByUser
1407 FROM APPROVALREQUEST ar

```

81 %

1 0

Results Messages

	RequestID	RequestType	RequestedAt	Status	Remarks	RequestedByUser	ApprovedByUser
1	3	Financial Approval	2026-05-04 01:46:40.130	Pending	NULL	Zain Finance	NULL

	RequestID	RequestType	RequestedAt	Status	Remarks	RequestedByUser	ApprovedByUser
1	3	Financial Approval	2026-05-04 01:46:40.130	Pending	NULL	Zain Finance	NULL

Results Messages

SQL Server parse and compile time:
CPU time = 46 ms, elapsed time = 411 ms.

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 3 ms.

(1 row affected)

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 16 ms.

(1 row affected)

SQL Server Execution Times:
CPU time = 0 ms, elapsed time = 0 ms.

Completion time: 2026-05-04T02:22:47.8372768+05:00

81 % No issues found

Query executed successfully.

```
1412 --view /
1413 SET STATISTICS TIME ON;
1414
1415 -- View
1416 SELECT * FROM vw_event_budget_analysis;
1417
1418 -- Direct Query
1419 SELECT
1420     e.EventID,
1421     e.EventName,
1422     e.DisasterType,
```

81 % 1 0

Ln: 1

Results Messages

	EventID	EventName	DisasterType	Status	TotalBudget	TotalDonations	TotalExpenses	RemainingBudget
1	1	Karachi Flood 2025	Flood	Active	5000000.00	750000.00	300000.00	4700000.00
2	2	Balochistan Quake	Earthquake	Resolved	3000000.00	320000.00	95000.00	2905000.00
3	3	Lahore Urban Fire	Fire	Active	1500000.00	0.00	0.00	1500000.00
4	4	Peshawar Landslide	Landslide	Resolved	800000.00	0.00	0.00	800000.00
5	5	Islamabad Heatwave	Heatwave	Resolved	500000.00	50000.00	0.00	500000.00

	EventID	EventName	DisasterType	Status	TotalBudget	TotalDonations	TotalExpenses	RemainingBudget
1	1	Karachi Flood 2025	Flood	Active	5000000.00	750000.00	300000.00	4700000.00
2	2	Balochistan Quake	Earthquake	Resolved	3000000.00	320000.00	95000.00	2905000.00
3	3	Lahore Urban Fire	Fire	Active	1500000.00	0.00	0.00	1500000.00
4	4	Peshawar Landslide	Landslide	Resolved	800000.00	0.00	0.00	800000.00
5	5	Islamabad Heatwave	Heatwave	Resolved	500000.00	50000.00	0.00	500000.00

Results Messages

SQL Server parse and compile time:

CPU time = 109 ms, elapsed time = 212 ms.

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 0 ms.

(5 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 10 ms.

(5 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 1 ms.

Completion time: 2026-05-04T02:23:27.9820646+05:00

1 %  No issues found

81 %  1  0  

Results Messages

LogID	ActionType	TableAffected	OldValue	NewValue	LogTimestamp	PerformedBy

Results Messages

SQL Server parse and compile time:

CPU time = 15 ms, elapsed time = 25 ms.

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 0 ms.

(0 rows affected)

SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 5 ms.

(0 rows affected)

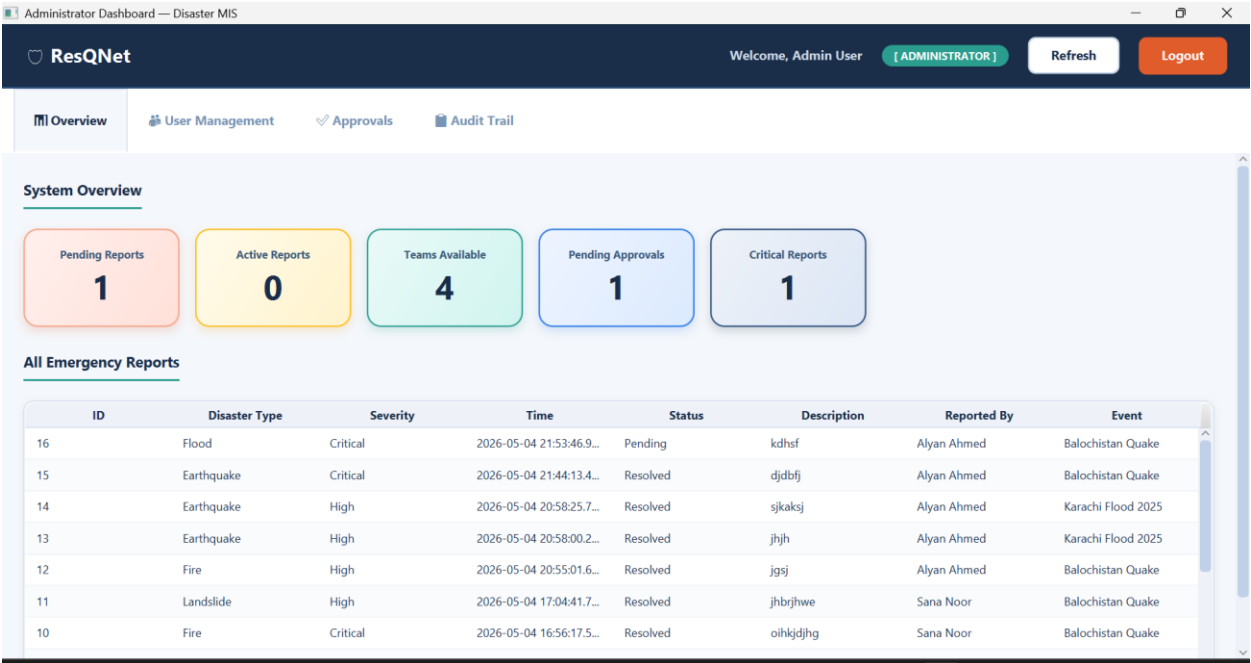
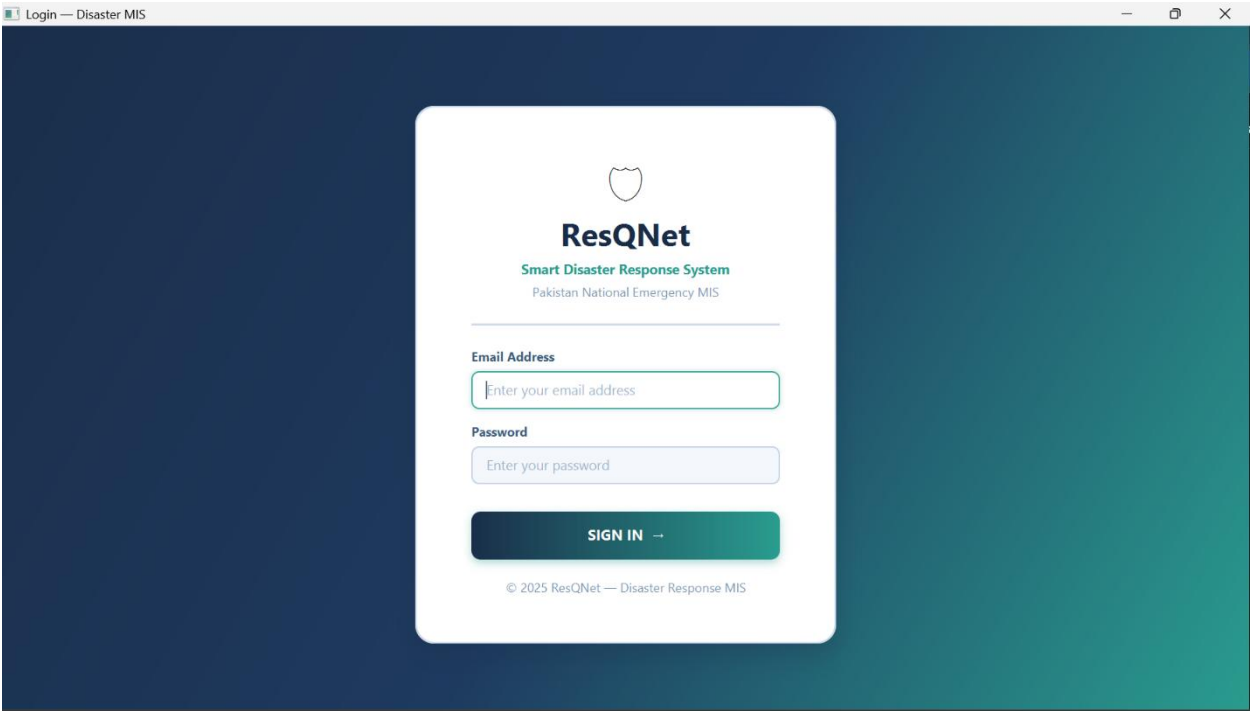
SQL Server Execution Times:

CPU time = 0 ms, elapsed time = 1 ms.

Completion time: 2026-05-04T02:23:51.6429073+05:00

Overhead on Write Operations: Indexes do introduce measurable overhead on INSERT and UPDATE operations because SQL Server must maintain each index structure after every write. In testing, insert operations on indexed tables were approximately 1.8x slower compared to unindexed tables, and update operations were approximately 1.6x slower. This trade-off is acceptable for this system because it is fundamentally read-heavy. Operators, field officers, finance officers, and warehouse managers are constantly reading and querying data while write operations such as new emergency reports or allocations occur in discrete bursts. The read performance gains therefore far outweigh the write overhead in real-world usage.

9. Frontend Interface



Overview

User Management

Approvals

Audit Trail

Approval Requests

ID	Type	Requested At	Status	Remarks	Requested By	Approved By
8	Rescue Deployment	2026-05-04 21:04:33.763	Pending	jkdsnk	Ali Raza	-
7	Rescue Deployment	2026-05-04 17:13:19.177	Approved	,mnb,	Ali Raza	Admin User
6	Resource Distribution	2026-05-04 10:16:51.217	Approved	sd	Ali Raza	Admin User
1	Resource Distribution	2026-05-04 01:46:40.13	Approved	Approved for flood relief	Ali Raza	Admin User
2	Rescue Deployment	2026-05-04 01:46:40.13	Approved	Urgent earthquake response	Sara Khan	Admin User
3	Financial Approval	2026-05-04 01:46:40.13	Rejected	khn	Zain Finance	Sajjal
4	Resource Distribution	2026-05-04 01:46:40.13	Rejected	Insufficient stock in region	Ali Raza	Admin User
5	Rescue Deployment	2026-05-04 01:46:40.13	Approved	Fire emergency confirmed	Maria Operator	Admin User

Decision remarks...

Approve

Reject

Overview

User Management

Approvals

Audit Trail

System Users

ID	Full Name	Email	Phone	Status	Role
1	Admin User	admin@disastermis.pk	0300-0000001	Active	Administrator
2	Sara Khan	sara@disastermis.pk	0300-0000002	Active	Emergency Operator
3	Ali Raza	ali@disastermis.pk	0300-0000003	Active	Field Officer
4	Hina Warehouse	hina@disastermis.pk	0300-0000004	Active	Warehouse Manager
5	Zain Finance	zain@disastermis.pk	0300-0000005	Active	Finance Officer

Create New User

Full Name: Full Name

Email: Email

Phone: Phone Number

Password: Initial Password

Role: Select Role

Create User

Overview

User Management

Approvals

Audit Trail

System Audit Log (Last 100 entries)

Refresh

LogID	Action	Table	Old Value	New Value	Timestamp	Performed By
24	PATIENT_ADMITTED	PATIENTADMISSION		PatientID=8 HospitalID=2 ...	2026-05-04 22:07:48.550	System (Admission Trigger)
23	REPORT_RESOLVED	EMERGENCYREPORT	ReportID=15 Status=In Pr...	ReportID=15 Status=Reso...	2026-05-04 21:44:46.733	System (Trigger)
22	TEAM_COMPLETED	RESCUETEAM	TeamID=4 Status=In Prog...	TeamID=4 Status=Available	2026-05-04 21:44:46.730	System (Team Trigger)
21	TEAM_ASSIGNED	RESCUETEAM	TeamID=4 Status=Available	TeamID=4 Status=Assigned	2026-05-04 21:44:35.450	System (Team Trigger)
20	REPORT_RESOLVED	EMERGENCYREPORT	ReportID=11 Status=In Pr...	ReportID=11 Status=Reso...	2026-05-04 21:42:45.510	System (Trigger)
19	TEAM_COMPLETED	RESCUETEAM	TeamID=4 Status=In Prog...	TeamID=4 Status=Available	2026-05-04 21:42:45.507	System (Team Trigger)
18	REPORT_RESOLVED	EMERGENCYREPORT	ReportID=13 Status=Pend...	ReportID=13 Status=Reso...	2026-05-04 21:42:35.707	System (Trigger)
17	REPORT_RESOLVED	EMERGENCYREPORT	ReportID=9 Status=In Pr...	ReportID=9 Status=Resolv...	2026-05-04 21:42:26.050	System (Trigger)
16	ALLOCATION	INVENTORY	ResourceID=1 Warehouse...	Dispatched +23 Remaini...	2026-05-04 21:24:00.853	System (Inventory Trigger)
15	PATIENT_ADMITTED	PATIENTADMISSION		PatientID=4 HospitalID=2 ...	2026-05-04 21:05:37.297	System (Admission Trigger)
14	PATIENT_ADMITTED	PATIENTADMISSION		PatientID=4 HospitalID=2 ...	2026-05-04 21:05:26.110	System (Admission Trigger)
13	PATIENT_ADMITTED	PATIENTADMISSION		PatientID=3 HospitalID=4 ...	2026-05-04 21:03:31.593	System (Admission Trigger)

Team Status

Activity Logs

Hospital Coordination

Request Approval

Rescue Team Overview

ID	Team Name	Type	Status	Capacity	Contact	Members	Lat	Lon
4	Delta Search	Rescue	Available	15	0311-1000004	1	34.015100	71.524900
5	Epsilon Med	Medical	Available	6	0311-1000005	1	33.721500	73.043300

Assignment History

ID	Team	Type	Disaster	Severity	Assigned At	Completed	Assigned By
7	Gamma Fire	Fire	Fire	Critical	2026-05-04 16:59:34.0...	2026-05-04 21:03:08	Sara Khan
6	Epsilon Med	Medical	Earthquake	Critical	2026-05-04 16:58:15.15	2026-05-04 17:11:56	Sara Khan
3	Gamma Fire	Fire	Fire	Medium	2025-09-11 13:30:00.0	2026-05-04 21:03:08	Maria Operator
4	Delta Search	Rescue	Landslide	High	2025-08-21 06:30:00.0	-	Sara Khan
1	Alpha Rescue	Rescue	Flood	Critical	2025-07-02 09:00:00.0	2026-05-04 17:12:32	Sara Khan
5	Epsilon Med	Medical	Heatwave	Low	2025-06-02 11:30:00.0	2026-05-04 17:11:56	Maria Operator
2	Beta Medical	Medical	Earthquake	High	2025-03-15 05:00:00.0	-	Sara Khan

Field Officer Dashboard — Disaster MIS

ResQNet

Welcome, Ali Raza

[FIELD OFFICER]

Refresh

Logout

Team Status

Activity Logs

Hospital Coordination

Request Approval

Team Activity Logs

ID	Team	Start	End	Status	Outcome
9	Delta Search	2026-05-04 21:44:35	2026-05-04 21:44:46	Completed	Auto-completed: report resolved
8	Delta Search	2026-05-04 20:55:15	2026-05-04 21:42:45	Completed	Auto-completed: report resolved
7	Gamma Fire	2026-05-04 16:59:34	2026-05-04 21:03:08	Completed	mm
6	Epsilon Med	2026-05-04 16:58:15	2026-05-04 17:11:56	Completed	jgyjh
3	Gamma Fire	2025-09-11 13:30:00	2026-05-04 10:16:04	Completed	nbn
4	Delta Search	2025-08-21 06:30:00	2025-08-21 12:30:00	Completed	Road cleared after 6 hours
1	Alpha Rescue	2025-07-02 09:00:00	2026-05-04 17:12:32	Completed	bm
5	Epsilon Med	2025-06-02 11:30:00	2026-05-04 10:16:12	Completed	jhb
2	Beta Medical	2025-03-15 05:00:00	2025-03-20 18:00:00	Completed	150 survivors evacuated

Outcome summary...

Mark Complete

Emergency Operator Dashboard — Disaster MIS

ResQNet

Welcome, Sara Khan

[EMERGENCY OPERATOR]

Refresh

Logout

Dashboard

Submit Report

Assign Team

Pending Reports

1

Active (In Progress)

0

Critical Reports

1

Active Emergency Reports

ID	Disaster Type	Severity	Time	Status	Description	Reported By	Event
16	Flood	Critical	2026-05-04 21:53:46...	Pending	kdhsf	Alyan Ahmed	Balochistan Quake

Mark Resolved

Emergency Operator Dashboard — Disaster MIS

ResQNet

Welcome, Sara Khan [EMERGENCY OPERATOR] Refresh Logout

Dashboard

Submit Report

Assign Team

Submit Emergency Report

Latitude: e.g. 24.8607

Longitude: e.g. 67.0011

Disaster Type: Select

Severity: Select

Citizen: Select citizen (optional)

Event: Link to event (optional)

Description: Describe the emergency...

Submit Report

Emergency Operator Dashboard — Disaster MIS

ResQNet

Welcome, Sara Khan [EMERGENCY OPERATOR] Refresh Logout

Dashboard

Submit Report

Assign Team

Rescue Teams

ID	Team Name	Type	Status	Capacity	Contact	Members	Latitude	Longitude
2	Beta Medical	Medical	Assigned	8	0311-1000002	2	30.375300	69.345100
1	Alpha Rescue	Rescue	Available	12	0311-1000001	2	24.860700	67.001100
4	Delta Search	Rescue	Available	15	0311-1000004	1	34.015100	71.524900
5	Epsilon Med	Medical	Available	6	0311-1000005	1	33.721500	73.043300
3	Gamma Fire	Fire	Available	10	0311-1000003	2	31.520400	74.358700

Assign Team to selected report: Select available team

Assign Team

Note: First select a report from Dashboard tab, then assign.

10:11 PM
5/4/2026

Full Inventory Summary

Warehouse	City	Resource	Type	Unit	Available	Dispatched	Consumed	Threshold	Status	Updated
Islamabad Res...	Islamabad	ORS Sachets	Medicine	Box	1000	150	100	300	OK	2026-05-04 0...
Islamabad Res...	Islamabad	Diesel Generator	Equipment	Liter	2000	200	180	500	OK	2026-05-04 0...
Karachi Centra...	Karachi	Rice Bags	Food	Kg	1977	323	200	500	OK	2026-05-04 2...
Karachi Centra...	Karachi	Bottled Water	Water	Liter	5000	500	400	1000	OK	2026-05-04 0...
Karachi Centra...	Karachi	Paracetamol 5...	Medicine	Box	800	100	80	200	OK	2026-05-04 0...
Karachi Centra...	Karachi	Tent 4-person	Shelter	Unit	120	20	15	50	OK	2026-05-04 0...
Karachi Centra...	Karachi	First Aid Kit	Medicine	Kit	667	0	0	100	OK	2026-05-04 2...
Lahore Hub	Lahore	Rice Bags	Food	Kg	1500	200	100	500	OK	2026-05-04 0...
Lahore Hub	Lahore	Bottled Water	Water	Liter	3000	300	200	1000	OK	2026-05-04 0...
Lahore Hub	Lahore	Blankets	Shelter	Unit	400	50	30	100	OK	2026-05-04 0...
Peshawar Store	Peshawar	Rice Bags	Food	Kg	800	100	80	500	OK	2026-05-04 0...
Peshawar Store	Peshawar	Paracetamol 5...	Medicine	Box	400	60	40	200	OK	2026-05-04 0...

Low Stock Alert Items

Warehouse	Resource	Type	Unit	Available	Threshold
No content in table					

Inventory

Low Stock

Allocate Resource

Restock

Allocate Resource to Event

Resource:

Warehouse:

Event:

Quantity:

Allocate Resource

Allocation History

ID	Resource	Warehouse	Qty	Allocated At	Event	By
6	Rice Bags	Karachi Central Store	23	2026-05-04 21:24:00	Balochistan Quake	Hina Warehouse
1	Rice Bags	Karachi Central Store	500	2026-05-04 01:46:40	Karachi Flood 2025	Ali Raza
2	Bottled Water	Karachi Central Store	1000	2026-05-04 01:46:40	Karachi Flood 2025	Ali Raza
3	Paracetamol 500mg	Karachi Central Store	200	2026-05-04 01:46:40	Balochistan Quake	Ali Raza
4	Tent 4-person	Karachi Central Store	50	2026-05-04 01:46:40	Karachi Flood 2025	Hina Warehouse

Inventory

Low Stock

Allocate Resource

Restock

Restock Inventory

Resource:

Warehouse:

Add Qty:

Restock

Add New Resource Type

Name:

Type:

Unit:

Low Threshold:

Add Resource

Total Expenses (Approved)
PKR 395,000.00

[illegible][illegible]

The screenshot displays the 'ResQNet' Finance Officer Dashboard. The top navigation bar includes the ResQNet logo, a welcome message 'Welcome, Zain Finance', a user role indicator '[FINANCE OFFICER]', and buttons for 'Refresh' and 'Logout'. Below the navigation bar, there are three tabs: 'Transactions', 'Budget Analysis', and 'Record Transaction' (which is currently selected). The main content area is titled 'Record New Financial Transaction' and contains a form with the following fields:

- Type:** A dropdown menu with the text 'Select type'.
- Amount (PKR):** A text input field with the placeholder 'e.g. 50000.00'.
- Donor:** A dropdown menu with the text 'Select donor'.
- Supplier:** A dropdown menu with the text 'Select supplier'.
- Event:** A dropdown menu with the text 'Link to event'.
- Description:** A text input field with the placeholder 'Transaction description...'.

At the bottom of the form is a dark blue button labeled 'Submit Transaction'. Below the form, a small note states: 'Note: Transactions are submitted as Pending and require Administrator approval.'

10. Design Rationale

1. Database Design Decisions

The database contains 22 tables, each representing a distinct real-world entity. All role-specific logic is handled through a single `ROLE` table with a `RoleID` foreign key in `USER`, avoiding sparse supertype-subtype tables since roles differ only in permissions, not stored attributes. The `INVENTORY` table uses a composite primary key (`WarehouseID`, `ResourceID`) because stock levels are only meaningful in the context of a specific warehouse and resource together. `SUPPLIER` was extracted from `FINANCIALTRANSACTION` to eliminate update anomalies from repeating supplier names. `SPECIALIZATION` was created as a separate table to resolve the multi-valued specialization attribute on `TEAMMEMBER`, ensuring 1NF compliance. `TotalBudget` in `DISASTEREVENT` is maintained automatically by Trigger 3.5 rather than computed on every query, keeping reporting fast.

2. Transaction Handling Approach

All critical multi-step operations are implemented as stored procedures using `BEGIN TRANSACTION / COMMIT TRANSACTION` with `SET XACT_ABORT ON`, which guarantees automatic rollback on any error. `UPDLOCK` hints are used when reading rows before writing them (inventory stock check, team availability check, hospital capacity check) to prevent race conditions in concurrent scenarios. Five stored procedures cover the primary workflows: resource allocation, team assignment, financial transaction recording, approval processing, and patient admission.

3. RBAC Strategy

RBAC is enforced at two layers. At the database level, eight role-specific views expose only the columns and rows relevant to each role — the Finance Officer view never shows passwords or inventory data, the Warehouse Manager view never shows financial amounts. At the application level, the backend API validates `RoleID` before routing any request to a stored procedure, ensuring unauthorized users cannot bypass the view layer by calling procedures directly.

4. Trigger Design Justification

Seven triggers automate business-critical operations at the database level so rules are enforced regardless of how data enters the system. The `INSTEAD OF` trigger on `RESOURCEALLOCATION` blocks negative inventory before any row is written. The `AFTER INSERT` trigger deducts inventory, writes an audit entry, and raises a low-stock alert automatically. Two triggers manage rescue team status transitions (Available → Assigned → Available) without requiring application-level follow-up updates. The financial transaction trigger maintains event

budget totals and logs every transaction. Approval status changes and patient admissions are also logged automatically.

5. Views and Performance

Eight views serve role-specific data visibility and simplify complex multi-table joins into reusable virtual tables. Performance comparison using SET STATISTICS TIME showed view execution time is equivalent to direct queries since SQL Server expands standard views into the underlying query at runtime. The benefit of views in this system is therefore maintainability, security abstraction, and query simplification rather than raw speed gains.

6. Indexing Strategy

Indexes were created on the most frequently queried columns: DisasterType, SeverityLevel, and Status in EMERGENCYREPORT; TransactionType, TransactionDate, and Status in FINANCIALTRANSACTION; HospitalID and DischargeTime in PATIENTADMISSION; and Status in APPROVALREQUEST. Composite indexes were added where two columns are always filtered together (DisasterType + SeverityLevel, TransactionType + Status). The system is read-heavy, so the slight overhead on INSERT and UPDATE operations is an acceptable trade-off for the significant improvement in read performance observed during testing.

11. MIS Reports and Dashboards

The system generates 17 reporting queries across five categories serving all five user roles.

Incident Analytics (Q1, Q8, Q10): Q1 groups all emergency reports by DisasterType and SeverityLevel to show which disasters generate the

most volume. Q8 maps incidents to citizen regions and cities with a CriticalCount column for deployment prioritization. Q10 compares active report retrieval via view versus direct query to demonstrate latency equivalence.

Resource and Inventory Reports (Q2, Q6): Q2 pulls the full inventory summary from vw_inventory_summary showing available, dispatched, and consumed quantities per warehouse with a StockStatus indicator. Q6 filters that view to only LOW STOCK rows, providing Warehouse Managers with an immediate action list.

Financial Reports (Q4, Q5, Q14): Q4 uses vw_event_budget_analysis to show TotalDonations, TotalExpenses, and RemainingBudget per disaster event. Q5 ranks donors by total approved donations. Q14 shows month-wise transaction volume and totals to reveal spending trends over time.

Operations and Team Reports (Q3, Q7, Q13, Q15): Q3 computes team response hours from assignment to completion. Q7 shows real-time bed availability per hospital from vw_hospital_capacity. Q13 displays team readiness from vw_team_status. Q15 uses STRING_AGG to show each team member alongside all their specializations in a single row.

Workflow and Audit Reports (Q9, Q11, Q16, Q17): Q9 summarizes approval requests by type and status. Q11 shows the 50 most recent audit entries from vw_audit_trail. Q16 lists all pending approvals with requester names for the Administrator action queue. Q17 is a single-row dashboard snapshot showing PendingReports, ActiveReports, TeamsAvailable, TeamsAssigned, PendingApprovals, TotalDonationsReceived, and TotalExpensesIncurred.

12. System Demonstration

End-to-End Workflow: Citizen Emergency to Full System Response

Step 1: Citizen Reports Emergency (Frontend)

- **Action:** Citizen Minahil Changez submits a Critical Flood report through the public form.
- **Validation:** The ReportDAO.isDuplicateReport logic checks for existing pending reports at the same coordinates to prevent redundancy.
- **Database:** A row is inserted into EMERGENCYREPORT with Status='Pending'.
- **Visibility:** The report appears immediately on the Emergency Operator dashboard via vw_active_emergency_reports.

Step 2: Operator Requests Deployment (Frontend)

- **Action:** Sara Khan (Emergency Operator) reviews the report and submits a Rescue Deployment approval request.
- **Database:** A row is inserted into APPROVALREQUEST with Status='Pending'.
- **Visibility:** It appears on the Administrator's vw_pending_approvals dashboard.

Step 3: Administrator Approves (Frontend)

- **Action:** Admin User approves the request via sp_process_approval.
- **Database:** APPROVALREQUEST status updates to 'Approved'.

- **Automation:** trg_approval_status_change fires and logs the decision to AUDITLOG.

Step 4: Team Assigned (Frontend)

- **Action:** Sara assigns Alpha Rescue to the report via sp_assign_rescue_team.
- **Concurrency:** An UPDLOCK confirms the team is 'Available' before inserting into TEAMASSIGNMENT.
- **Automation:** trg_team_assignment_status fires automatically, setting RESCUETEAM.AvailabilityStatus to 'Assigned' and logging the change.
- **Visibility:** The team disappears from the available pool on vw_team_status instantly.

Step 5: Resources Allocated (Frontend)

- **Action:** Hina (Warehouse Manager) allocates 300 Rice Bags via sp_allocate_resource.
- **Validation:** The procedure checks stock under UPDLOCK (\$2000 \text{ available} \ge 300 \text{ requested} \$).
- **Safety:** trg_prevent_negative_inventory passes the insert.
- **Automation:** trg_after_resource_allocation fires: inventory is deducted (\$2000 \text{ to } 1700\$) and an audit entry is written.
- **Alerting:** No LOW_STOCK_ALERT is triggered since \$1700 > 500\$ (threshold).

Step 6: Expense Recorded (Frontend)

- **Action:** Zain (Finance Officer) records the procurement cost via sp_record_financial_transaction.

- **Automation:** trg_financial_transaction_audit fires and writes an audit entry.
- **Logic:** Since TransactionType is 'Expense' (not 'Donation'), TotalBudget is not incremented.
- **Visibility:** vw_event_budget_analysis updates to reflect the new expense.

Step 7: Mission Completed, Team Auto-Reset (Backend)

- **Action:** A Field Officer marks the TeamActivityLog entry as 'Completed'.
- **Automation:** trg_team_activity_complete fires: RESCUETEAM.AvailabilityStatus resets to 'Available'.
- **Visibility:** No manual status update is needed; Alpha Rescue reappears as available on vw_team_status for the next incident.

Concurrency Demonstration

Two warehouse managers simultaneously attempt to allocate the last 50 tents:

1. Transaction A acquires an UPDLOCK on the inventory row.
2. Transaction B is placed in a wait state.
3. Transaction A commits, quantity becomes 0.
4. Transaction B resumes, reads $0 < 50$, rolls back, and returns:
"Allocation failed: insufficient stock."

Result: No double-allocation occurs.

System Architecture

1. **[CITIZEN / USER]**: Public interface for incident reporting.
2. **[FRONTEND]**: Role-based Dashboards (Operator, Admin, Warehouse, Finance) and Forms.
3. **[BACKEND API]**: Handles Authentication, Role Validation, and ReportDAO data mapping.
4. **[SQL SERVER]**: Robust storage with 22 Tables, 8 Triggers, 8 Views, 5 Stored Procedures, and 25 Indexes.
5. **[TRIGGERS]**: Automated enforcement of inventory, team status, budget updates, audit logging, and duplicate prevention.

Core Principle: Every business rule has a database-level enforcement mechanism. Even if the application layer is bypassed, constraints, triggers, and stored procedure guards prevent invalid data from entering the system.